

Physics Extended Essay

Investigation on the frequencies of oscillations of a coupled physical pendulum

Word count: 3890

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1) Introduction

1.1) Coupled physical pendulums

Simple harmonic motion refers to the motion of a body when the restoring force (on the body) is directly proportional to the displacement from equilibrium¹. Such conditions are met in a physical pendulum, which is a rigid body that undergoes fixed axis rotation about a fixed point².

A two-mass coupled pendulum is formed when two identical pendulums are attached by a soft spring³. The spring in between is assumed to obey Hooke's law (i.e., an ideal spring), which states that the displacement or size of the deformation (of the spring) is directly proportional to the deforming force or load⁴.

1.2) Modes of oscillation

A single oscillator has a single natural frequency. However, when two or more oscillators are coupled with springs, they have several natural frequencies and that the general motion (of the oscillator) is a combination of variations at all different natural frequencies⁵

In the case of a oscillations involving two coupled oscillators, the mode of oscillation is divided into two normal modes of oscillation, which are modes at which the masses can oscillate sinusoidally, and the modes which the masses oscillate depend on the initial conditions of oscillation. The pendula will oscillate in the 1st normal mode of oscillation ω_1 , in which the pendula solely oscillates in the 1st normal angular frequency, when the masses are initially displaced in phase. While the pendula will oscillate in the 2nd normal mode of

¹ Reference: Ford, A. L. Young, H. D. Freedman, R. A. *Sears and Zemansky's University Physics With Modern Physics* (13th edition). Pearson

² Reference: Peter D. (20th July, 2022). *Physical pendulum*. Physics LibreTexts.

[https://phys.libretexts.org/Bookshelves/Classical_Mechanics/Classical_Mechanics_\(Dourmashkin\)/24%3A_Physical_Pendulums/24.02%3A_Physical_Pendulum](https://phys.libretexts.org/Bookshelves/Classical_Mechanics/Classical_Mechanics_(Dourmashkin)/24%3A_Physical_Pendulums/24.02%3A_Physical_Pendulum)

³ Reference: *Coupled Pendulum* [Online] UC Santa Cruz UCSC PHYSICS DEMONSTRATION ROOM.

<https://ucscphysicsdemo.sites.ucsc.edu/physics-5a6a/coupled-pendulum/>

⁴ Reference: *Elasticity* [Online] Britannica. <https://www.britannica.com/science/elasticity-physics>

⁵ Reference: Taylor, John R. (John Robert), 1939-. (2005). *Classical mechanics*. Sausalito, Calif. University Science Books

oscillation ω_2 , in which the pendula solely oscillates in the 2nd normal mode angular frequency, when the masses are initially displaced in antiphase.

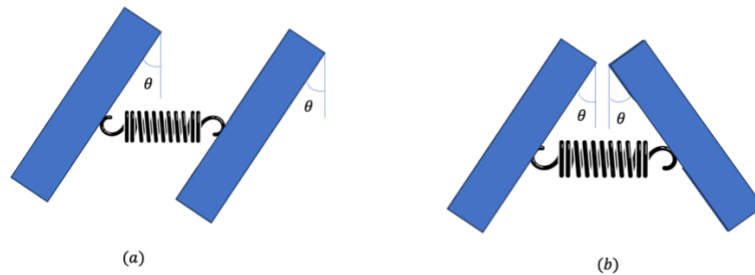


Figure 1. illustration of the (a) 1st normal mode when the masses are displaced in phase and (b) the 2nd normal mode when they are displaced in antiphase

The initial displacements of the pendula determine the resulting waveform of its motion. A special initial condition, called resonance, occurs when one pendulum stays at rest and the other is placed in an arbitrary initial displacement (shown in *Figure 2.*)

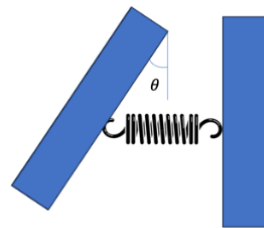


Figure 2. Illustration of pendula displaced in resonance

When the pendula are displaced in resonance, the pendula will oscillate in linear combinations of ω_1 and ω_2 , which are known as the average ω_{avg} and beat ω_{beat} angular frequencies, which are given by,⁶

$$\omega_{avg} = \frac{\omega_1 + \omega_2}{2}$$

$$\omega_{beat} = \frac{\omega_2 - \omega_1}{2}$$

⁶ *Coupled pendulum* (N.D.) New York University.
https://physics.nyu.edu/~physlab/GenPhysII/PhysIII/Intermediate_Physics_I_writeups/Coupled-Pendulum_08-10-017.pdf

1.3) Research question

Coupled pendulums not only frequently appear in clock machineries, and the mechanism of coupled pendulums are also present in molecular vibrations, in which bonds within a molecule are distorted in the same way as springs are stretched and compressed.

The bond strength between different diatomic molecules vary, which in turn affects the atoms' frequency of vibration. I am interested in investigating such a situation but examining molecular vibration is out of my reach. Instead, such a situation can be simulated classically, by varying the coupling strength between two masses in a coupled pendulum, which can be done so by changing the position of the spring from the pivot.

In addition, upon the various possibilities of sending the coupled pendulum in oscillation, I would like to particularly investigate on the oscillation of pendulums in resonance as it exhibits the superposition of the 2 normal modes of oscillation, allowing me to obtain both ω_1 and ω_2 , and thus, ω_{avg} and ω_{beat} in one go, rather than conducting the experiment in both in-phase and out-of-phase measurement. Therefore, I would like to investigate **“In a coupled physical pendulum with pendula oscillating in resonance, how does distance of spring from pivot affect the 1st and 2nd normal mode angular frequencies, hence, the average and beat angular frequencies?”**

2) Qualitative prediction

As the distance of the spring from the pivot increases, the torque exerting on the arms of the pendula increases. The greater the torque acting on the pendula, the greater the ω_2 . However, the ω_1 will remain unchanged since it is independent of the spring extension. As ω_{avg} and ω_{beat} is a linear combination of the 2 normal mode frequencies, an increases distance of the spring from the pivot would result in an increase in the magnitude of both ω_{avg} and ω_{beat} .

3) Hypothesis

As the distance of the spring from the pivot increases, ω_1 remains unchanged and ω_2 increases. Hence, ω_{avg} and ω_{Beat} increases accordingly.

4) Variable table

Dependent variable		Average and beat angular frequencies of oscillation
Controlled variable	Independent variable	Distance of the spring from the pivot
	Fixed variables	1. The masses of the pendula 2. The lengths of the pendula 3. The length of the spring 4. The spring constant of the spring 5. The height of the pendula 6. The initial separation between the pendula 7. The type of rotary motion sensor
Uncontrolled variable		1. Friction acting on the pendula. 2. Air resistance acting on the pendula.

5) Methodology

5.1) Experimental set-up

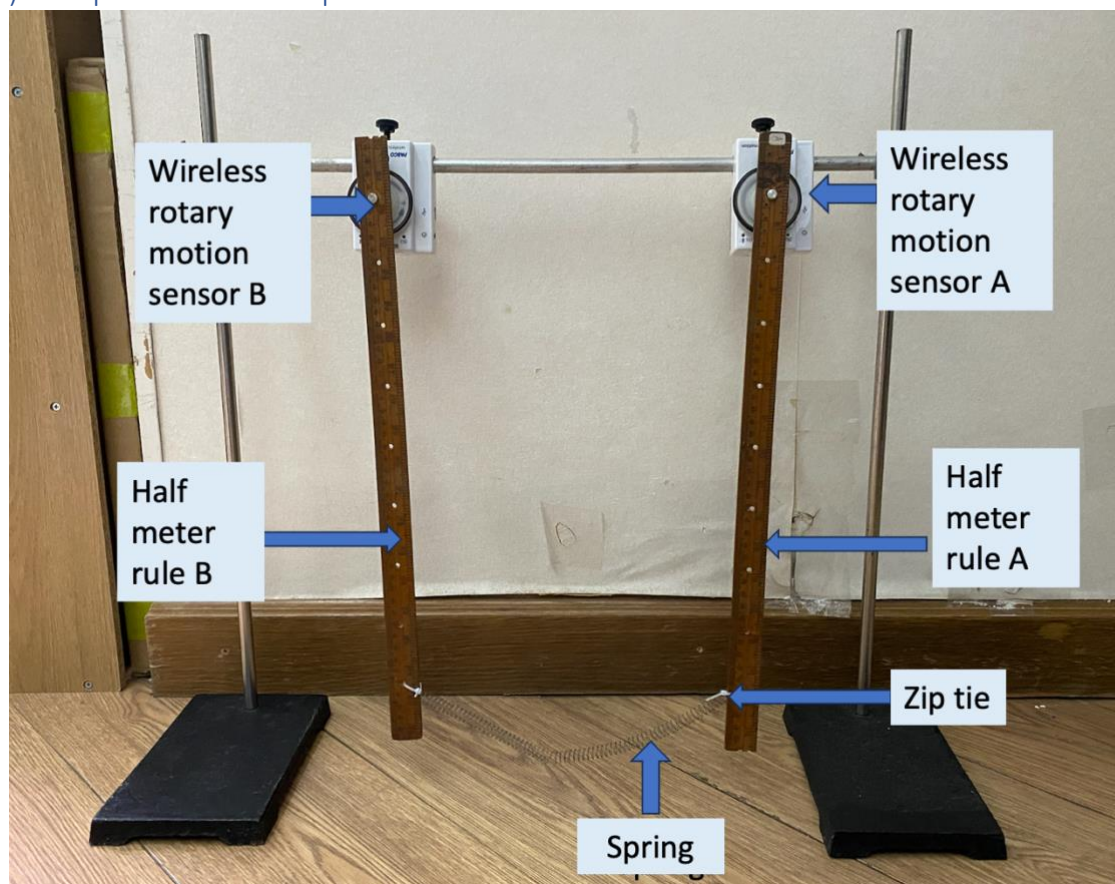


Figure 3. Labelled image of experimental set-up

5.2) Apparatus

Half meter rule (with holes) x 2	PASCO wireless rotary motion sensor x 2	Stand and clamp x2
Metal rod x 1	Spring x 1	Zip ties
Laptop with PASCO Capstone installed	Electronic balance x 1	

6) Method of data collection

To collect data, two different methods were attempted. The first method was to obtain the average and beat angular frequencies by obtaining their respective periods of oscillation.

Graphs of angle (in radians) against time were plotted on a data-logger. The time required for one individual complete oscillation was taken as the period of the average frequency, while the time required for one complete oscillation of the sinusoidal wave envelope was taken as half the period of the beat frequency. The equation for the conversion of period T into angular frequency ω , which is $\omega = \frac{2\pi}{T}$ was used to obtain the angular frequencies of oscillation. (Refer to **Figure 4**)

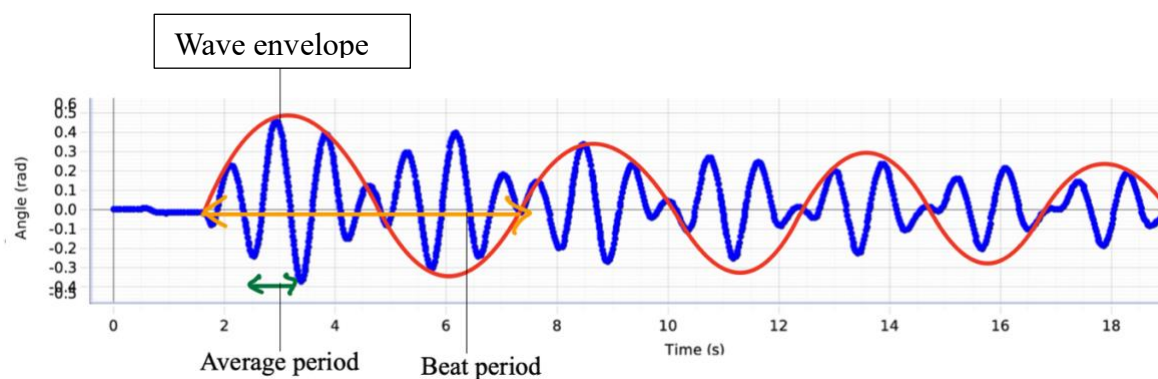


Figure 4. Angular displacement against time graph when $l = 0.15\text{m}$

However, the experimental results showed that as the distance of the spring from the pivot is increased, the angular displacement-time graph lost its sinusoidal envelope waveform, thus, the value for ω_{beat} could not be obtained (as shown in **Figure 5**). Therefore, this method was abandoned.

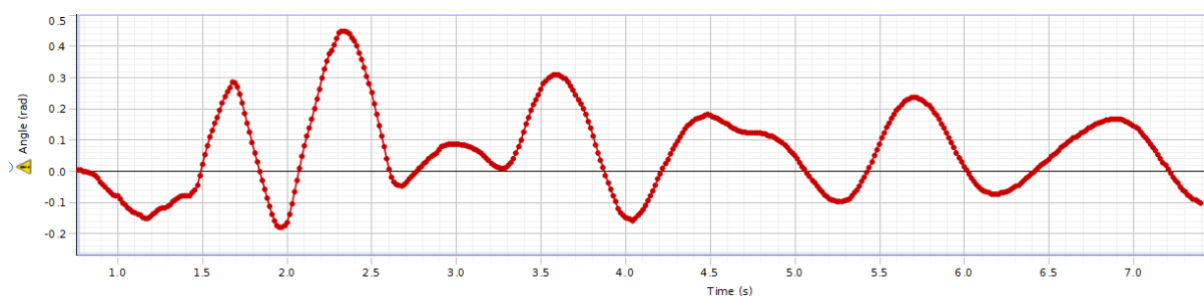


Figure 5. Angular displacement against time graph when $l = 0.4\text{m}$.

Another approach to data collection was to use Fast Fourier Transform (FFT) to obtain the 2 angular frequencies of oscillation. Utilizing Pasco Capstone's FFT display, the 2 peaks in the FFT spectrum of the greatest amplitude were taken as the angular frequencies of oscillation (refer to *Figure 6*).

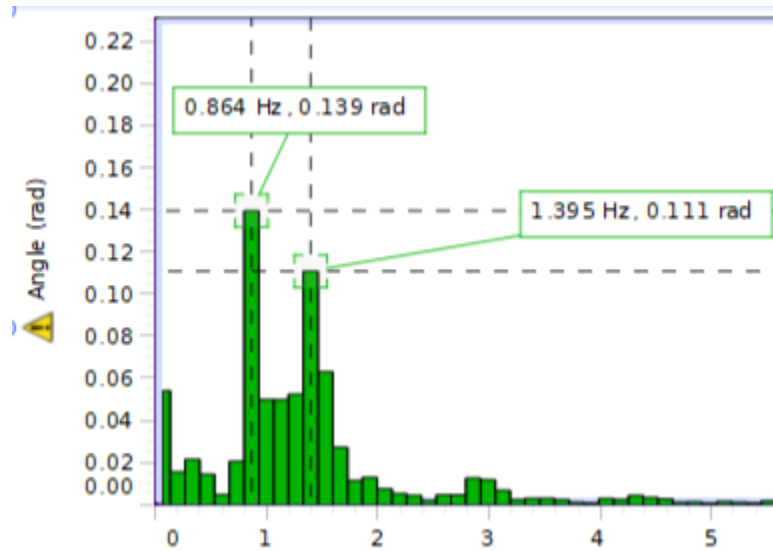


Figure 6. Fast Fourier transform (FFT) spectrum obtained in trial session.

The peak that corresponds to a higher frequency gives the value of ω_2 , while that of the lower frequency gives the value of the ω_1 . Thus, the angular frequencies of oscillation can be obtained when the spring is connected to different distances from the pivot of the pendulum.

7) Experimental procedures

7.1) Determination of spring constant

1. Attach the spring on a stand and clamp and measure its equilibrium length.
2. Hang 20g slotted masses on the lower end of the spring and record the final length of the spring.
3. Deduct the equilibrium length of the spring from its final length to obtain the spring extension x .
4. Repeat step 2 and 3 for 5 more times (with a total of 6 slotted masses).
5. Obtain the weight W of the slotted masses using $W = mg$, where m is the total mass of the slotted masses in kg.
6. Plot a graph of weight W against extension Δx , apply a linear line of best fit and plot the worst lines.
7. The slope of the line of best fit is the spring constant of the spring, and the uncertainty of the slope can be found by $\Delta m = \frac{MAX(m) - MIN(m)}{2}$, where $MAX(m)$ is the slope of the maximum worst line and $MIN(m)$ is the slope of the minimum worst line

7.2) Finding the first and second normal mode frequencies

1. Create an FFT spectrum and an angle-time graph of one of the pendulums on the PASCO Capstone data-logging interface.
2. Set the horizontal axis of the FFT spectrum to a maximum of 2.0Hz.
3. Set the maximum number of bins to 4096 in FFT properties.
4. Set the sample rate of the rotary motion sensor to 40Hz.
5. Keep one of the pendulums at rest and displace the other to an arbitrary small angle from equilibrium.
6. Press “Start recording”. After 20 seconds, press “stop recording.”
7. Use the coordinate tools on the FFT spectrum, find the 2 highest peaks on the graph to obtain the frequencies of the first and second normal mode of oscillation.

8) Data

8.1) Raw data

8.1.1) Basic measurements

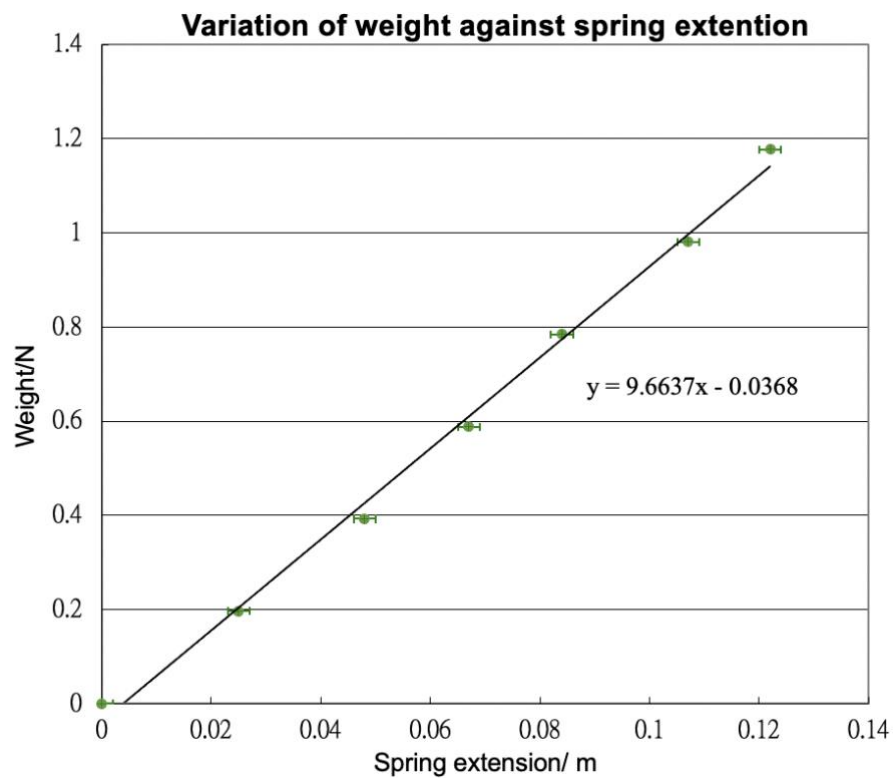
Length of half meter rule l_0/m	0.500 ± 0.001
Mass of the ruler M/g	45.55 ± 0.01
Spring constant k/Nm^{-1}	9.6637 ± 0.0000
Moment of inertia I/kgm^2	0.02271 ± 0.00002

Table 1. Basic measurements and quantities

Determination of the spring constant k and Δk

W/N	$\Delta W/\text{N}$	x/m	$\Delta x/\text{m}$
0.0000	0.0000	0.000	0.002
0.1962	0.0001	0.025	0.002
0.3924	0.0001	0.048	0.002
0.5886	0.0001	0.067	0.002
0.7848	0.0001	0.084	0.002
0.9810	0.0001	0.107	0.002
1.1772	0.0001	0.122	0.002

Table 2. Weight of slotted mass(es) and spring extension



Graph 1. Weight against spring extension

Calculation of ΔW

$$\Delta W = g\Delta m$$

Sample calculation:

$$\Delta W = (9.81) \left(\frac{0.01}{1000} \right) = 0.0000981 \approx 0.00001$$

As the magnitude of the vertical error bars are too small, worst lines were not drawn such that the spring constant is assumed to be error-free.

Therefore, the spring constant of the spring is $9.6637 Nm^{-1}$.

Determination of the moment of inertia and its uncertainty

I of the pendulum:

$$I = \frac{1}{12} M(l_0^2 + w^2) + MD^2$$

Where D is the distance between the midpoint of the ruler and the pivot, and w is the thickness of the half meter rule.

Calculation of ΔI

$$\Delta I = \frac{1}{12} M l_0^2 \left[\frac{\Delta M}{M} + 2 \left(\frac{\Delta l_0}{l_0} \right) \right] + \frac{1}{12} M w^2 \left[\frac{\Delta M}{M} + 2 \left(\frac{\Delta w}{w} \right) \right] + MD^2 \left[\frac{\Delta M}{M} + 2 \left(\frac{\Delta D}{D} \right) \right]$$

Sample calculation:

$$I = \frac{1}{12} [(0.5)^2 + (0.026)^2] + \left(\frac{45.55}{1000} \right) (0.20)^2 \approx 0.0227 \text{ kg m}^2$$

$$\Delta I = \frac{1}{12} \left(\frac{45.55}{1000} \right) (0.5)^2 \left[\frac{(0.01)}{(45.55)} + 2 \left(\frac{0.001}{0.5} \right) \right] + \frac{1}{12} \left(\frac{45.55}{1000} \right) (0.026)^2 \left[\frac{0.01}{45.55} + 2 \left(\frac{0.001}{0.026} \right) \right] +$$

$$\left(\frac{45.55}{1000} \right) (0.20)^2 \left[\frac{0.01}{45.55} + 2 \left(\frac{0.001}{0.20} \right) \right] \approx \pm 2 \times 10^{-5} \text{ kg m}^2$$

8.1.2) Frequency measurements

$l \pm \Delta l/m$ $\pm 0.001m$	$f_1 \pm \Delta f_1/Hz$ ($\pm 0.001Hz$)					$\overline{f_1}/Hz$	$\overline{\Delta f_1}/Hz$
	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5		
0.400	0.302	0.302	0.302	0.302	0.302	0.302	0.001
0.350	0.302	0.302	0.302	0.302	0.302	0.302	0.001
0.300	0.302	0.302	0.302	0.302	0.302	0.302	0.001
0.250	0.482	0.302	0.302	0.302	0.302	0.302	0.001
0.200	0.302	0.302	0.302	0.302	0.302	0.302	0.001
0.150	0.302	0.302	0.302	0.302	0.302	0.302	0.001
0.100	0.302	0.302	0.302	0.285	0.302	0.302	0.001
0.050	0.302	0.302	0.302	0.302	0.302	0.302	0.001

Table 3. Raw data table for $f_1 \pm \Delta f_1$

$l \pm \Delta l/m$ $\pm 0.001m$	$f_2 \pm \Delta f_2/Hz$ ($\pm 0.001Hz$)					$\overline{f_2}/Hz$	$\overline{\Delta f_2}/Hz$
	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5		
0.400	1.770	1.770	1.770	1.648	1.648	1.72	0.07
0.350	1.634	1.634	1.634	1.604	1.604	1.62	0.03
0.300	1.404	1.404	1.404	1.404	1.404	1.404	0.001
0.250	1.282	1.282	1.282	1.282	1.282	1.282	0.001
0.200	0.929	0.966	0.966	0.929	0.929	0.94	0.02
0.150	0.798	0.798	0.764	0.764	0.798	0.78	0.02
0.100	0.576	0.554	0.576	0.554	0.576	0.57	0.01
0.050	0.395	0.395	0.395	0.402	0.402	0.398	0.004

Table 4. Raw data table for $f_2 \pm \Delta f_2$

Calculation of $\overline{f_1}$ and $\overline{f_2}$

$$\overline{f_1} = \frac{\text{Trial 1 } (f_0) + \text{Trial 2 } (f_0) + \text{Trial 3 } (f_0) + \text{Trial 4 } (f_0) + \text{Trial 5 } (f_0)}{5}$$

$$\overline{f_2} = \frac{\text{Trial 1 } (f_1) + \text{Trial 2 } (f_1) + \text{Trial 3 } (f_1) + \text{Trial 4 } (f_1) + \text{Trial 5 } (f_1)}{5}$$

Calculation of $\overline{\Delta f_1}$ and $\overline{\Delta f_2}$

1. Taking the average of the individual scale uncertainties of f_0

$$\overline{\Delta f_1} = \frac{\text{Trial 1 } (\Delta f_0) + \text{Trial 2 } (\Delta f_0) + \text{Trial 3 } (\Delta f_0) + \text{Trial 4 } (\Delta f_0) + \text{Trial 5 } (\Delta f_0)}{5}$$

$$\overline{\Delta f_2} = \frac{\text{Trial 1 } (\Delta f_1) + \text{Trial 2 } (\Delta f_1) + \text{Trial 3 } (\Delta f_1) + \text{Trial 4 } (\Delta f_1) + \text{Trial 5 } (\Delta f_1)}{5}$$

2. Half of range obtained from 5 trials.

$$\Delta \bar{f}_1 = \frac{MAX(f_0) - MIN(f_0)}{2}$$

$$\Delta \bar{f}_2 = \frac{MAX(f_1) - MIN(f_1)}{2}$$

3. Standard deviation

$$\Delta \bar{f}_1 = \sqrt{\frac{[Trial\ 1\ (f_0) - \bar{f}_0]^2 + [Trial\ 2\ (f_0) - \bar{f}_0]^2 + [Trial\ 3\ (f_0) - \bar{f}_0]^2 + [Trial\ 4\ (f_0) - \bar{f}_0]^2 + [Trial\ 5\ (f_0) - \bar{f}_0]^2}{5}}$$

$$\Delta \bar{f}_2 = \sqrt{\frac{[Trial\ 1\ (f_1) - \bar{f}_1]^2 + [Trial\ 2\ (f_1) - \bar{f}_1]^2 + [Trial\ 3\ (f_1) - \bar{f}_1]^2 + [Trial\ 4\ (f_1) - \bar{f}_1]^2 + [Trial\ 5\ (f_1) - \bar{f}_1]^2}{5}}$$

The method which propagates a larger uncertainty would be used to obtain the uncertainty.

Sample calculation:

$\bar{f}_1$ and $\bar{f}_2$ when $l = 0.40m$

$$\bar{f}_1 = \frac{0.302 + 0.302 + 0.302 + 0.302 + 0.302}{5} = 0.302Hz$$

$$\bar{f}_2 = \frac{1.770 + 1.770 + 1.770 + 1.648 + 1.648}{5} = 1.721Hz$$

$\Delta \bar{f}_1$ and $\Delta \bar{f}_2$ when $l = 0.40m$

$$\Delta \bar{f}_1 = \frac{0.001 + 0.001 + 0.001 + 0.001 + 0.001}{5} = \pm 0.001Hz$$

$$\Delta \bar{f}_2 = \frac{0.001 + 0.001 + 0.001 + 0.001 + 0.001}{5} = \pm 0.001Hz$$

$$\Delta \bar{f}_1 = \frac{0.302 - 0.302}{2} = \pm 0.000Hz$$

$$\Delta \bar{f}_2 = \frac{0.770 - 0.648}{2} \approx \pm 0.06Hz$$

$$\Delta \bar{f}_1 = \sqrt{\frac{+(0.302-0.302)^2+(0.302-0.302)^2+(0.302-0.302)^2+(0.302-0.302)^2+(0.302-0.302)^2}{5}} \approx \pm 0.0Hz$$

$$\Delta \bar{f}_2 = \sqrt{\frac{(1.770-1.720)^2+(1.770-1.942)^2+(1.770-1.942)^2+(1.648-1.942)^2+(1.648-1.942)^2}{5}} \approx \pm 0.07Hz$$

Therefore, $\bar{f}_1 = (0.302 \pm 0.001)Hz$ and $\bar{f}_2 = (1.72 \pm 0.07) Hz$

l/m ($\pm 0.001m$)	$\bar{\omega}_1 / rads^{-1}$	$\Delta\bar{\omega}_1 / rads^{-1}$	$\bar{\omega}_2 / rads^{-1}$	$\Delta\bar{\omega}_2 / rads^{-1}$
0.400	1.898	0.006	10.8	0.4
0.350	1.898	0.006	10.2	0.2
0.300	1.898	0.006	8.821	0.006
0.250	1.898	0.006	8.055	0.006
0.200	1.898	0.006	5.9	0.1
0.150	1.898	0.006	4.9	0.1
0.100	1.898	0.006	3.56	0.06
0.050	1.898	0.006	2.50	0.03

Table 5. Processed data table for $\bar{\omega}_1 \pm \Delta\bar{\omega}_1$ and $\bar{\omega}_2 \pm \Delta\bar{\omega}_2$

Calculation of $\bar{\omega}_1$ and $\bar{\omega}_2$:

$$\bar{\omega}_1 = 2\pi\bar{f}_1$$

$$\bar{\omega}_2 = 2\pi\bar{f}_2$$

Calculation of $\Delta\bar{\omega}_1$ and $\Delta\bar{\omega}_2$:

$$\Delta\bar{\omega}_1 = \left(\frac{\Delta\bar{f}_1}{\bar{f}_1} \right) \bar{\omega}_1$$

$$\Delta\bar{\omega}_2 = \left(\frac{\Delta\bar{f}_2}{\bar{f}_2} \right) \bar{\omega}_2$$

Sample calculation:

$\bar{\omega}_1$ and $\bar{\omega}_2$ when $l = 0.4m$,

$$\bar{\omega}_1 = 2\pi(0.793) \approx 4.983 \text{ rads}^{-1}$$

$$\bar{\omega}_2 = 2\pi(1.924) \approx 12.1 \text{ rads}^{-1}$$

$\Delta\bar{\omega}_1$ and $\Delta\bar{\omega}_2$ when $l = 0.4m$,

$$\Delta\bar{\omega}_0 = \left(\frac{0.001}{0.302} \right) (1.898) \approx \pm 0.006 \text{ rads}^{-1}$$

$$\Delta\bar{\omega}_1 = \left(\frac{0.1}{1.720} \right) (10.8) \approx \pm 0.4 \text{ rads}^{-1}$$

9) Theoretical model

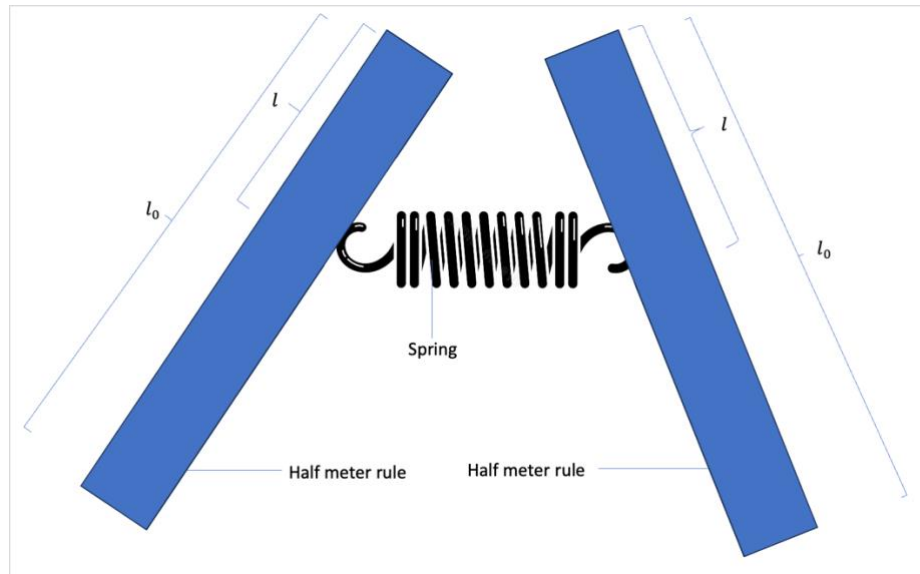


Figure 7. Schematic diagram showing the experimental set-up

A relationship between the angular frequencies of oscillation and the distance of the spring from the pivot shall be derived in order to theoretically model the motion of the pendula.

Consider the free body diagram of the coupled physical pendulum shown below:

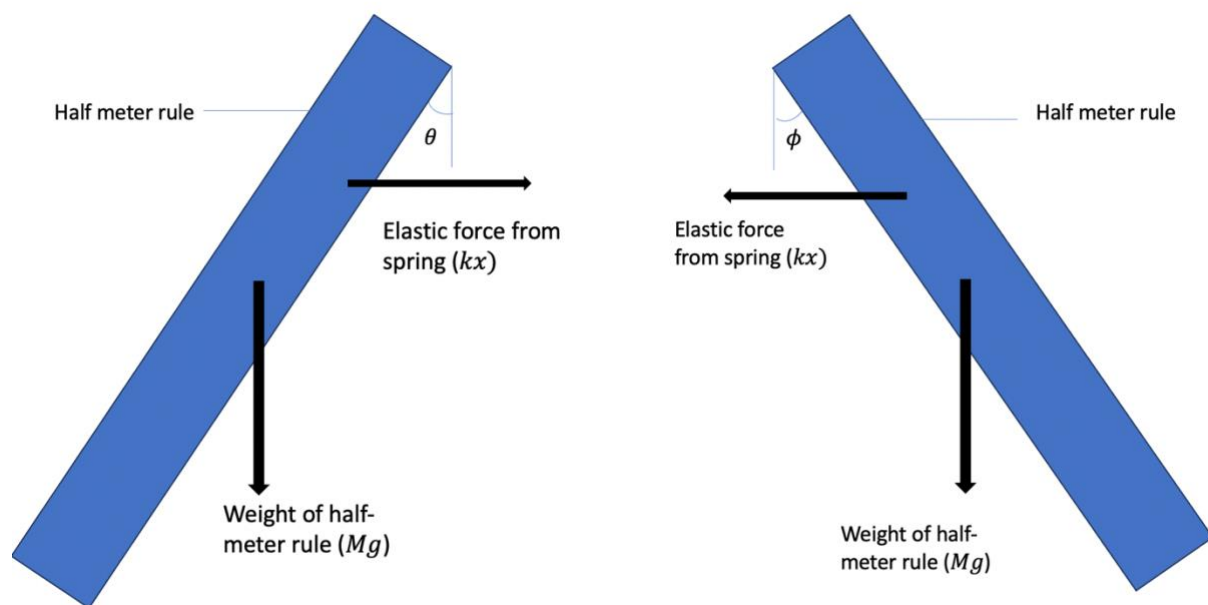


Figure 8. Free body diagrams of the arms of couple physical pendulum

When one pendulum is displaced θ from equilibrium while the other is displaced by ϕ , the extension of the spring x is given by,

$$x = l(\sin\theta - \sin\phi) \quad (1)$$

Where l is the distance between the spring and the pivot, and l_0 is the length between the brass mass and the pivot. The torque τ of the pendula are given by,

$$\tau = I\alpha \quad (2)$$

Where I is the moment of inertia and α is the angular acceleration. The restoring force applied by the spring on the pendulum is given by,

$$F = -kx \quad (3)$$

In which k is the spring constant and x being the extension of the spring. Defining M as the mass of the half meter rule, assuming the mass distribution along it is uniform and that the mass of both pendula are equal. By expressing $\alpha = \ddot{\theta}$, the torque about the pivot of the left pendulum is given by,

$$-lkx \cos\theta - \frac{Mgl_0}{2} \sin\theta = I\ddot{\theta} \quad (4)$$

The torque about the pivot of the right pendulum is given by,

$$lkx \cos\phi - \frac{Mgl_0}{2} \sin\phi = I\ddot{\phi} \quad (5)$$

Assuming that the initial angle of release is small, we can apply small angle approximation, in which, $\sin\theta \approx \theta$, $\cos\theta \approx 1$, $\sin\phi \approx \phi$ and $\cos\phi \approx 1$. The extension of string now becomes,

$$x = l(\theta - \phi) \quad (6)$$

The approximated torque about the pivot of the left pendulum is given by,

$$-lkx - \frac{Mgl_0}{2} \theta = I\ddot{\theta} \quad (7)$$

The approximated torque about the pivot of the right pendulum is given by,

$$lkx - \frac{Mgl_0}{2}\phi = I\ddot{\phi} \quad (8)$$

Adding (7) and (8) yields,

$$I(\ddot{\theta} + \ddot{\phi}) + \left(\frac{Mgl_0}{2}\right)(\theta + \phi) = 0$$

$$(\ddot{\theta} + \ddot{\phi}) + \frac{Mgl_0}{2I}(\theta + \phi) = 0 \quad (9)$$

Subtracting (8) from (7) yields,

$$I(\ddot{\theta} - \ddot{\phi}) = -lkx - \frac{Mgl_0}{2}\theta - lkx + \frac{Mgl_0}{2}\phi$$

$$I(\ddot{\theta} - \ddot{\phi}) + \left(\frac{Mgl_0}{2}\right)(\theta - \phi) + 2kxl = 0 \quad (10)$$

Substituting $x = l(\theta - \phi)$ and regrouping the terms,

$$I(\ddot{\theta} - \ddot{\phi}) + \left(\frac{Mgl_0}{2}\right)(\theta - \phi) + 2l^2k(\theta - \phi) = 0$$

$$I(\ddot{\theta} - \ddot{\phi}) + \left(\frac{Mgl_0}{2}\right)(\theta - \phi) + 2l^2k(\theta - \phi) = 0$$

$$(\ddot{\theta} - \ddot{\phi}) + \left(\frac{Mgl_0}{2I} + \frac{2kl^2}{I}\right)(\theta - \phi) = 0 \quad (11)$$

Therefore, the 2 linear second order homogenous differential equations are obtained:

$$(\ddot{\theta} + \ddot{\phi}) + \frac{Mgl_0}{2I}(\theta + \phi) = 0$$

$$(\ddot{\theta} - \ddot{\phi}) + \left(\frac{Mgl_0}{2I} + \frac{2kl^2}{I}\right)(\theta - \phi) = 0$$

Combining the 2 second order differential equation into a compact matrix form of $M\ddot{x} +$

$Kx = 0$, the matrix equation below would be obtained.

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \ddot{\theta} + \ddot{\phi} \\ \ddot{\theta} - \ddot{\phi} \end{bmatrix} + \begin{bmatrix} \frac{Mgl_0}{2I} & 0 \\ 0 & \frac{Mgl_0}{2I} + \frac{2kl^2}{I} \end{bmatrix} \begin{bmatrix} \theta + \phi \\ \theta - \phi \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad (12)$$

Solving the characteristic equation of this matrix equation,

$$\det(K - \omega^2 M) = 0 \quad (13)$$

In which $K = \begin{bmatrix} \frac{Mgl_0}{2I} & 0 \\ 0 & \frac{Mgl_0}{2I} + \frac{2kl^2}{I} \end{bmatrix}$ and $M = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$. Substituting the above matrices into the characteristic equation,

$$\det\left(\begin{bmatrix} \frac{Mgl_0}{2I} & 0 \\ 0 & \frac{Mgl_0}{2I} + \frac{2kl^2}{I} \end{bmatrix} - \omega^2 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right) = 0 \quad (14)$$

$$\left(\frac{Mgl_0}{2I} - \omega^2\right)\left(\frac{Mgl_0}{2I} + \frac{2kl^2}{I} - \omega^2\right) = 0$$

$$\omega_1^2 = \frac{Mgl_0}{2I} \text{ or } \omega_2^2 = \frac{Mgl_0}{2I} + \frac{2kl^2}{I}$$

$$\omega_1 = \sqrt{\frac{Mgl_0}{2I}} \text{ or } \omega_2 = \sqrt{\frac{Mgl_0}{2I} + \frac{2kl^2}{I}} \quad (15)$$

It is worth noting that for a coupled pendulum system, $\omega_2 > \omega_1$.

The general solution to (9) is,

$$\theta + \phi = A \cos(\omega_1 t) \quad (16)$$

The general solution to (11) is,

$$\theta - \phi = B \cos(\omega_2 t) \quad (17)$$

Solving (16) and (17) simultaneously,

$$\theta = \frac{A}{2} \cos(\omega_1 t) + \frac{B}{2} \cos(\omega_2 t) \quad (18)$$

$$\phi = \frac{A}{2} \cos(\omega_1 t) - \frac{B}{2} \cos(\omega_2 t) \quad (19)$$

Using the trigonometric sum to product formulas and assuming that the amplitudes A and B are equal,

$$\theta(t) = A \cos\left(\frac{\omega_1 + \omega_2}{2}t\right) \cos\left(\frac{\omega_1 - \omega_2}{2}t\right) \quad (20)$$

$$\phi(t) = -A \sin\left(\frac{\omega_1 + \omega_2}{2}t\right) \sin\left(\frac{\omega_1 - \omega_2}{2}t\right) \quad (21)$$

in which $\omega_{avg} = \frac{(\omega_0 + \omega_1)}{2}$, which is the angular frequency of each individual oscillation.

While $\omega_{beat} = \frac{(\omega_1 - \omega_0)}{2}$, which is the angular frequency which the sinusoidal wave envelop

oscillates with.⁷ Thus, substituting (15) into $\omega_{avg} = \frac{(\omega_0 + \omega_1)}{2}$ and $\omega_{beat} = \frac{(\omega_1 - \omega_0)}{2}$,

$$\omega_{avg} = \frac{1}{2} \sqrt{\frac{Mgl_0}{2I}} + \frac{1}{2} \sqrt{\frac{Mgl_0}{2I} + \frac{2kl^2}{I}} \quad (22)$$

$$\omega_{beat} = \frac{1}{2} \sqrt{\frac{Mgl_0}{2I} + \frac{2kl^2}{I}} - \frac{1}{2} \sqrt{\frac{Mgl_0}{2I}} \quad (23)$$

Hence, the average and beat angular frequency, which are the dependent variables, can be expressed as a function of l , the independent variable. Therefore, this allows us to determine the relationship between the angular frequencies of oscillation and the distance of the spring from the pivot.

⁷ Reference: EX-5563 Coupled pendula [Online] PASCO. https://cdn.pasco.com/lab_experiment/I_1224/EX-5563_Coupled_Pendula.pdf

10) Preliminary Processed data

10.2.1) Preliminary frequency data

According to the theoretical predictions made above,

$$\omega_1 = \sqrt{\frac{Mgl_0}{2I}} \text{ or } \omega_2 = \sqrt{\frac{Mgl_0}{2I} + \frac{2kl^2}{I}} \quad (15)$$

Taking the square on both sides of the formulas above

$$\omega_1^2 = \frac{Mgl_0}{2I} \text{ or } \omega_2^2 = \frac{Mgl_0}{2I} + \frac{2kl^2}{I} \quad (24)$$

Theoretically, referring to (24), as ω_1 is independent of the distance of the spring from the pivot l , a graph of ω_1 against l would result in a horizontal straight line, while a graph of ω_2^2 against l^2 would result in a linear graph. Displaying the experimental results graphically:

l/m ($\pm 0.001\text{m}$)	$\bar{\omega}_1 / \text{rads}^{-1}$	$\Delta\bar{\omega}_1 / \text{rads}^{-1}$
0.400	1.898	0.006
0.350	1.898	0.006
0.300	1.898	0.006
0.250	1.898	0.006
0.200	1.898	0.006
0.150	1.898	0.006
0.100	1.898	0.006
0.050	1.898	0.006

Table 6. Table of l and $\bar{\omega}_0$

$l^2 \pm \Delta(l^2)/\text{m}^2$	$\bar{\omega}_2^2 / \text{rads}^{-1}$	$\Delta(\bar{\omega}_2^2) / \text{rads}^{-1}$
0.1600 ± 0.0008	116.64	9
0.1225 ± 0.0007	104.04	4
0.0900 ± 0.0006	77.810	0.1
0.0625 ± 0.0005	64.88	0.1
0.0400 ± 0.0004	34.81	1
0.0225 ± 0.0003	24.01	1
0.0100 ± 0.0002	12.67	0.4
0.0025 ± 0.0001	6.25	0.2

Table 7. Table of l^2 and $\bar{\omega}_1^2$

Calculation of $\Delta(\omega_2^2)$

$$\Delta(\omega_2^2) = 2\omega_2\Delta\omega_2$$

Sample calculation

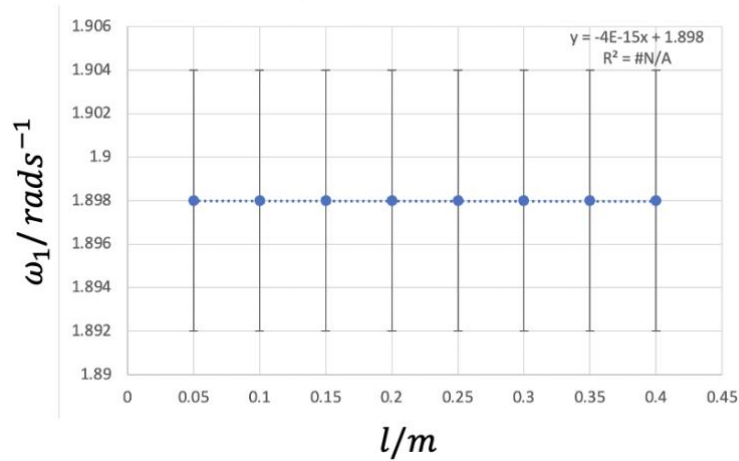
$\Delta(\omega_2^2)$ when $l = 0.4\text{m}$,

$$\Delta(\omega_2^2) = 2(10.8)(0.4) = \pm 8.64 \text{ rad}^2\text{s}^{-2} \approx 9 \text{ rad}^2\text{s}^{-2}$$

Therefore, $\omega_2^2 \pm \Delta(\omega_2^2) = (117 \pm 9) \text{ rad}^2\text{s}^{-2}$

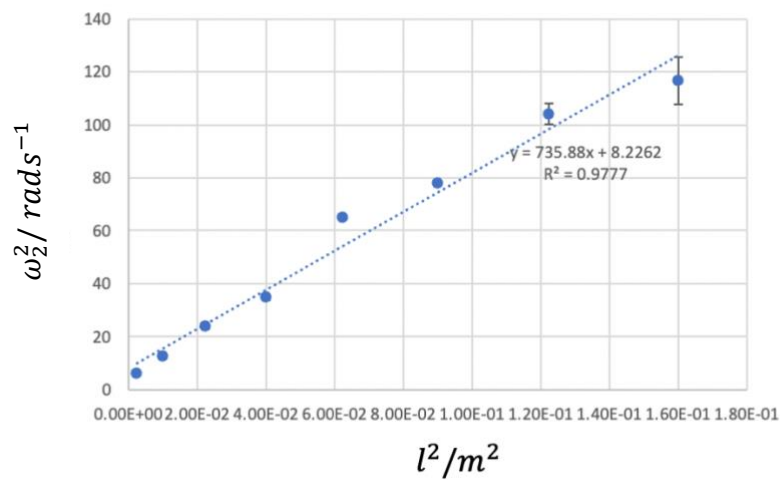
10.2.2) Preliminary frequency plot

Graph of ω_1 against l



Graph 2. Graphical relationship between ω_1 and l

Graph of ω_2^2 against l



Graph 3. Graphical relationship between ω_2^2 and l^2

As the error bars of most of the data points are too small, worst lines cannot be drawn.

Therefore, making use the LINEST function and taking reference to **Graph 2** and **Graph 3.**,

$$\omega_1 = 4.983 \pm 0.006 \quad (25)$$

$$\omega_2^2 = (735 \pm 50) l^2 + (8 \pm 4) \quad (26)$$

10.3) Further processed data

10.3.1) Further frequency data

l/m ($\pm 0.001m$)	$\overline{\omega_{avg}} / rads^{-1}$	$\Delta\overline{\omega_{avg}} / rads^{-1}$	$\overline{\omega_{Beat}} / rads^{-1}$	$\Delta\overline{\omega_{Beat}} / rads^{-1}$
0.400	6.349	0.4	4.451	0.4
0.350	6.049	0.2	4.151	0.2
0.300	5.3595	0.01	3.461	0.01
0.250	4.9765	0.01	3.0785	0.01
0.200	3.899	0.1	2.001	0.1
0.150	3.399	0.1	1.501	0.1
0.100	2.729	0.07	0.831	0.07
0.050	2.199	0.04	0.301	0.04

Table 8. Processed data for higher and lower frequencies

The average and beat frequency.

$$\overline{\omega_{avg}} = \frac{\overline{\omega_1} + \overline{\omega_2}}{2}$$

$$\overline{\omega_{Beat}} = \frac{\overline{\omega_2} - \overline{\omega_1}}{2}$$

The uncertainty of the average and beat frequency.

$$\Delta\overline{\omega_{avg}} = \Delta\overline{\omega_1} + \Delta\overline{\omega_2}$$

$$\Delta\overline{\omega_{Beat}} = \Delta\overline{\omega_1} + \Delta\overline{\omega_2}$$

Sample calculation:

$\overline{\omega_{avg}}$ and $\overline{\omega_{Beat}}$ when $l = 0.4m$,

$$\overline{\omega_{avg}} = \frac{1.898 + 10.8}{2} = 6.349 \text{ rads}^{-1}$$

$$\overline{\omega_{Beat}} = \frac{10.8 - 1.898}{2} = 4.451 \text{ rads}^{-1}$$

$\Delta\overline{\omega_{avg}}$ and $\Delta\overline{\omega_{Beat}}$ when $l = 0.4\text{m}$,

$$\Delta\overline{\omega_{avg}} = 0.006 + 0.4 \approx \pm 0.4 \text{ rads}^{-1}$$

$$\Delta\overline{\omega_{Beat}} = 0.006 + 0.4 \approx \pm 0.4 \text{ rads}^{-1}$$

According to the theoretical predictions made above,

$$\omega_{avg} = \frac{1}{2}\sqrt{\frac{Mgl_0}{2I}} + \frac{1}{2}\sqrt{\frac{Mgl_0}{2I} + \frac{2kl^2}{I}} \quad (22)$$

$$\omega_{Beat} = \frac{1}{2}\sqrt{\frac{Mgl_0}{2I} + \frac{2kl^2}{I}} - \frac{1}{2}\sqrt{\frac{Mgl_0}{2I}} \quad (23)$$

Rearranging the terms in (22) and (23),

$$(\omega_{avg} - \frac{1}{2}\sqrt{\frac{Mgl_0}{2I}})^2 = \frac{Mgl_0}{8I} + \left(\frac{k}{2I}\right)l^2 \quad (27)$$

$$(\omega_{Beat} + \frac{1}{2}\sqrt{\frac{Mgl_0}{2I}})^2 = \frac{Mgl_0}{8I} + \left(\frac{k}{2I}\right)l^2 \quad (28)$$

Theoretically, plotting a graph of $(\omega_{avg} - \frac{1}{2}\sqrt{\frac{Mgl_0}{2I}})^2$ against l^2 and a graph of $(\omega_{Beat} +$

$\frac{1}{2}\sqrt{\frac{Mgl_0}{2I}})^2$ against l^2 would result in a linear graph.

10.3.2) Final relation

$l^2 \pm \Delta(l^2)/\text{m}^2$	$(\omega_{avg} - \frac{1}{2}\sqrt{\frac{Mgl_0}{2I}})^2/\text{rad}^2\text{s}^{-2}$	$\Delta \left[(\omega_{avg} - \frac{1}{2}\sqrt{\frac{Mgl_0}{2I}})^2 \right] / \text{rad}^2\text{s}^{-2}$
0.1600 ± 0.0008	27	4
0.1225 ± 0.0007	24	2
0.0900 ± 0.0006	18.1	0.1
0.0625 ± 0.0005	15.0	0.1
0.0400 ± 0.0004	7.8	0.6
0.0225 ± 0.0003	5.2	0.5
0.0100 ± 0.0002	2.6	0.2
0.0025 ± 0.0001	1.19	0.09

Table 9. Processed data for $(\omega_{avg} - \frac{1}{2}\sqrt{\frac{Mgl_0}{2I}})^2$ and l^2

$l^2 \pm \Delta(l^2)/m^2$	$(\omega_{Beat} + \frac{1}{2}\sqrt{\frac{Mgl_0}{2I}})^2 / rad^2 s^{-2}$	$\Delta \left[(\omega_{Beat} + \frac{1}{2}\sqrt{\frac{Mgl_0}{2I}})^2 \right] / rad^2 s^{-2}$
0.1600 ± 0.0008	31	4
0.1225 ± 0.0007	28	2
0.0900 ± 0.0006	20.9	0.1
0.0625 ± 0.0005	17.5	0.1
0.0400 ± 0.0004	9.7	0.6
0.0225 ± 0.0003	6.8	0.5
0.0100 ± 0.0002	3.8	0.3
0.0025 ± 0.0001	2.0	0.1

Table 10. Processed data for $(\omega_{Beat} + \frac{1}{2}\sqrt{\frac{Mgl_0}{2I}})^2$ and l^2

Calculation of $\Delta(\omega_{avg} - \frac{1}{2}\sqrt{\frac{Mgl_0}{2I}})^2$

$$\Delta \left[\left(\omega_{avg} - \frac{1}{2}\sqrt{\frac{Mgl_0}{2I}} \right)^2 \right] = 2 \left(\omega_{avg} - \frac{1}{2}\sqrt{\frac{Mgl_0}{2I}} \right) \left[\Delta \omega_{avg} + \frac{1}{2}\sqrt{\frac{Mgl_0}{2I}} \left(\frac{\Delta M}{M} + \frac{\Delta l_0}{l_0} + \frac{\Delta I}{I} \right) \right]$$

Calculation of $\Delta(\omega_{Beat} + \frac{1}{2}\sqrt{\frac{Mgl_0}{2I}})^2$

$$\Delta \left[\left(\omega_{Beat} + \frac{1}{2}\sqrt{\frac{Mgl_0}{2I}} \right)^2 \right] = 2 \left(\omega_{Beat} + \frac{1}{2}\sqrt{\frac{Mgl_0}{2I}} \right) \left[\Delta \omega_{Beat} + \frac{1}{2}\sqrt{\frac{Mgl_0}{2I}} \left(\frac{\Delta M}{M} + \frac{\Delta l_0}{l_0} + \frac{\Delta I}{I} \right) \right]$$

Sample calculation:

$(\omega_{avg} - \frac{1}{2}\sqrt{\frac{Mgl_0}{2I}})^2$ and $(\omega_{Beat} + \frac{1}{2}\sqrt{\frac{Mgl_0}{2I}})^2$ when $l = 0.4m$,

$$(\omega_{avg} - \frac{1}{2}\sqrt{\frac{Mgl_0}{2I}})^2 = \left(6.349 - \frac{1}{2}\sqrt{\frac{(\frac{45.55}{1000})(9.81)(0.5)}{2(0.0227)}} \right)^2 \approx 27 \text{ rad}^2 s^{-2}$$

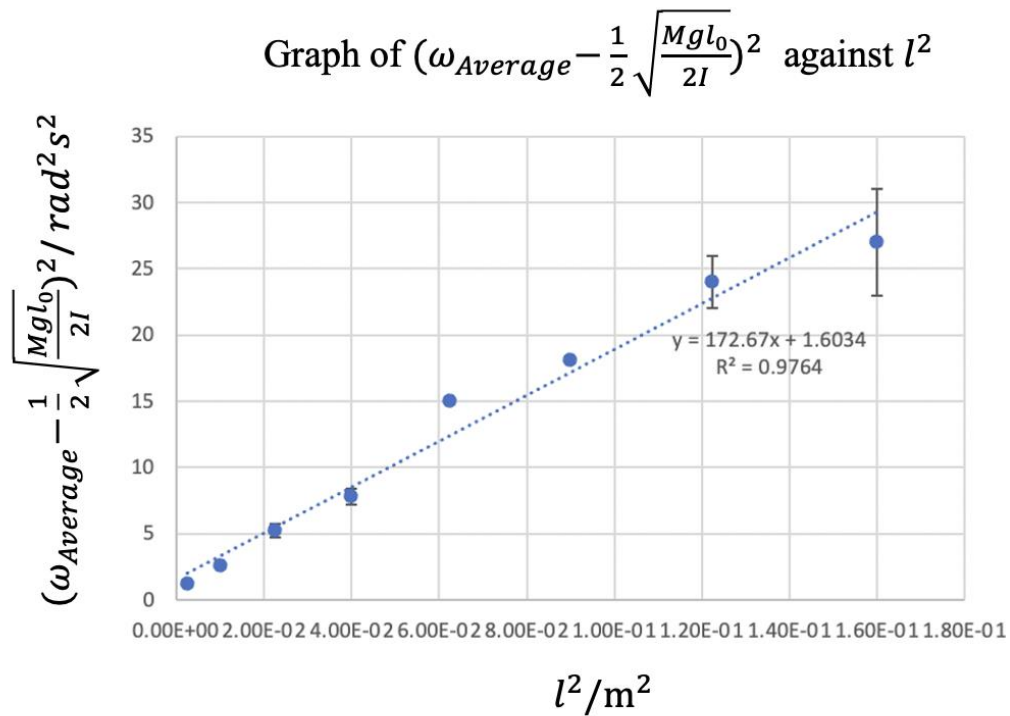
$$(\omega_{Beat} + \frac{1}{2}\sqrt{\frac{Mgl_0}{2I}})^2 = \left(4.451 + \frac{1}{2}\sqrt{\frac{(\frac{45.55}{1000})(9.81)(0.5)}{2(0.0227)}} \right)^2 \approx 31 \text{ rad}^2 s^{-2}$$

$\Delta \left[\left(\omega_{avg} - \frac{1}{2}\sqrt{\frac{Mgl_0}{2I}} \right)^2 \right]$ and $\Delta \left[\left(\omega_{Beat} + \frac{1}{2}\sqrt{\frac{Mgl_0}{2I}} \right)^2 \right]$ when $l = 0.4m$,

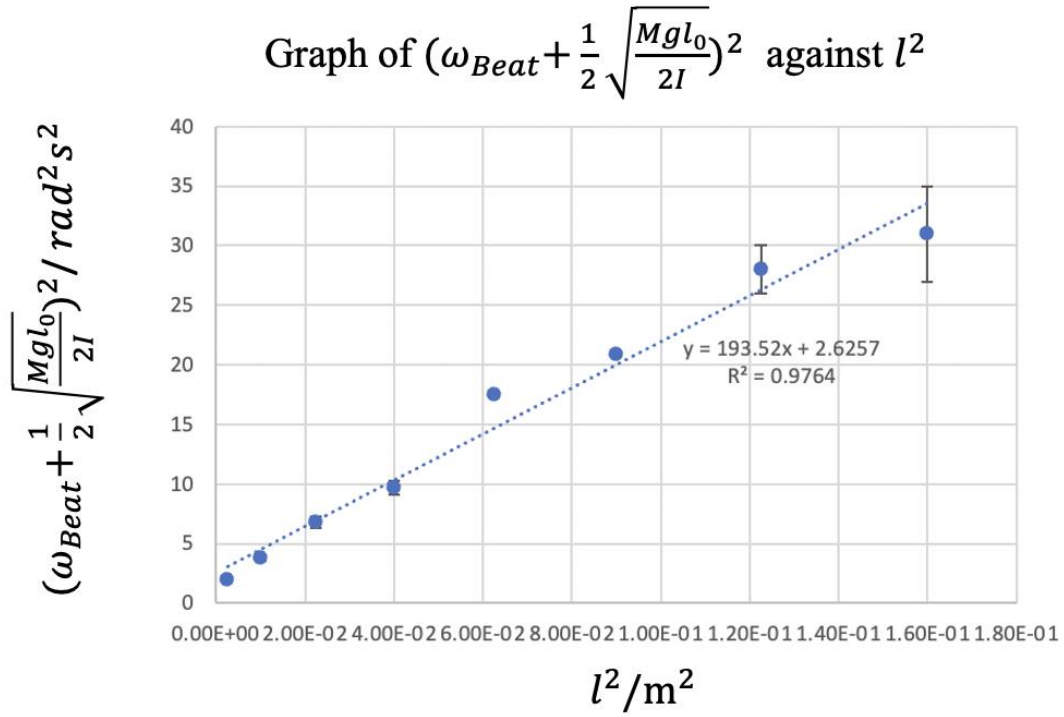
$$\Delta \left[\left(\omega_{avg} - \frac{1}{2}\sqrt{\frac{Mgl_0}{2I}} \right)^2 \right]$$

$$\begin{aligned}
&= 2 \left(6.349 - \frac{1}{2} \sqrt{\frac{\left(\frac{45.55}{1000}\right)(9.81)(0.5)}{2(0.0227)}} \right) \left[0.4 + \frac{1}{2} \sqrt{\frac{\left(\frac{45.55}{1000}\right)(9.81)(0.5)}{2(0.0227)}} \left(\frac{0.01}{45.55} + \frac{0.001}{0.500} + \frac{0.00002}{0.02271} \right) \right] \\
&\approx \pm 4 \text{ rad}^2 \text{ s}^{-2} \\
&\Delta \left[\left(\omega_{Beat} + \frac{1}{2} \sqrt{\frac{Mgl_0}{2I}} \right)^2 \right] \\
&= 2 \left(4.451 + \frac{1}{2} \sqrt{\frac{\left(\frac{45.55}{1000}\right)(9.81)(0.5)}{2(0.0227)}} \right) \left[0.4 + \frac{1}{2} \sqrt{\frac{\left(\frac{45.55}{1000}\right)(9.81)(0.5)}{2(0.0227)}} \left(\frac{0.01}{45.55} + \frac{0.001}{0.500} + \frac{0.00002}{0.02271} \right) \right] \\
&\approx \pm 4 \text{ rad}^2 \text{ s}^{-2}
\end{aligned}$$

10.3.3) Final results



Graph 4. Graph of $\left(\omega_{avg} - \frac{1}{2} \sqrt{\frac{Mgl_0}{2I}} \right)^2$ against l^2



Graph 5. Graph of $(\omega_{Beat} + \frac{1}{2}\sqrt{\frac{Mgl_0}{2I}})^2$ against l^2

As the error bars of most data points are too small, the LINEST function had to be used to determine the uncertainty of the slope and y-intercept of the graphs. Taking reference to **Graph 4** and **Graph 5** respectively,

$$(\omega_{avg} - \frac{1}{2}\sqrt{\frac{Mgl_0}{2I}})^2 = (172 \pm 11) l^2 + (1.6 \pm 0.9) \quad (29)$$

$$(\omega_{Beat} + \frac{1}{2}\sqrt{\frac{Mgl_0}{2I}})^2 = (194 \pm 12) l^2 + (3 \pm 1) \quad (30)$$

11) Discussion and conclusion

11.1) Discussion

11.1.1) Preliminary theoretical results

l/m ($\pm 0.001m$)	$\overline{\omega_1}/rads^{-1}$	$\Delta\overline{\omega_1}/rads^{-1}$
0.400	2.218	0.0007
0.350	2.218	0.0007
0.300	2.218	0.0007
0.250	2.218	0.0007
0.200	2.218	0.0007
0.150	2.218	0.0007
0.100	2.218	0.0007
0.050	2.218	0.0007

Table 11. Table of theoretical l and $\overline{\omega_0}$

$l^2 \pm \Delta(l^2)/m^2$	$\overline{\omega_2^2}/rads^{-1}$	$\Delta(\overline{\omega_2^2})/rads^{-1}$
0.1600 ± 0.0008	141.1	0.8
0.1225 ± 0.0007	109.2	0.7
0.0900 ± 0.0006	81.5	0.6
0.0625 ± 0.0005	58.1	0.5
0.0400 ± 0.0004	39.0	0.4
0.0225 ± 0.0003	24.1	0.3
0.0100 ± 0.0002	13.4	0.2
0.0025 ± 0.0001	7.05	0.1

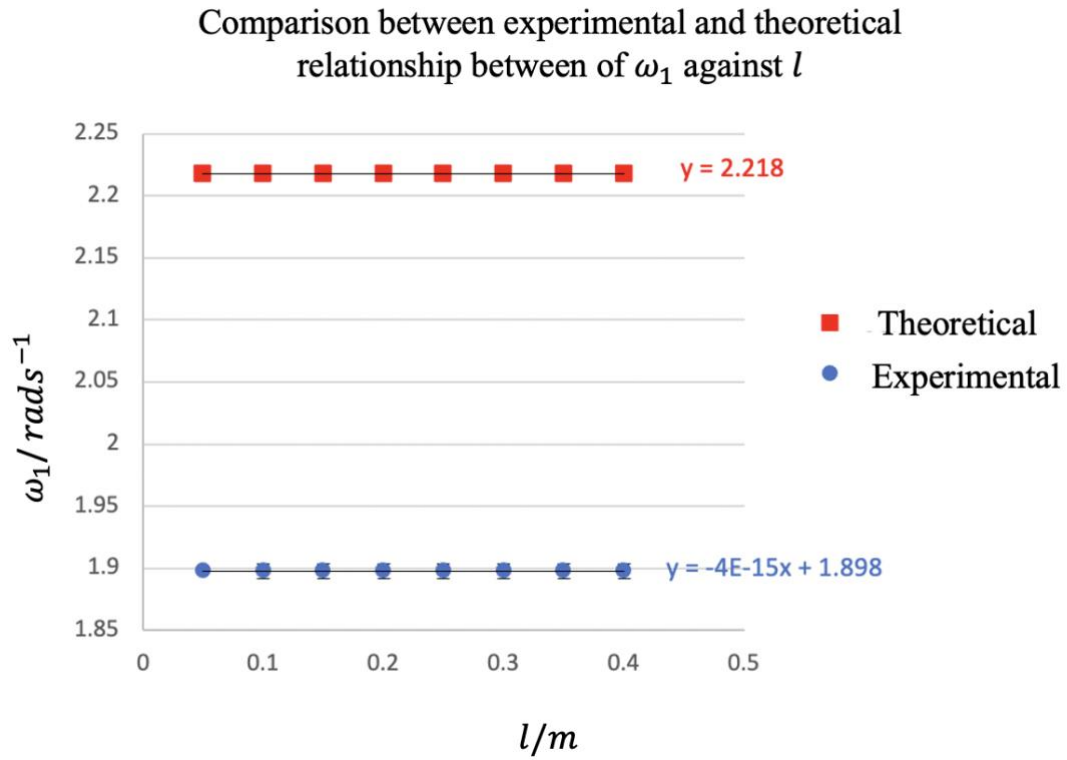
Table 12. Table of theoretical l^2 and $\overline{\omega_1^2}$

11.1.2) Preliminary comparison between experimental and theoretical results

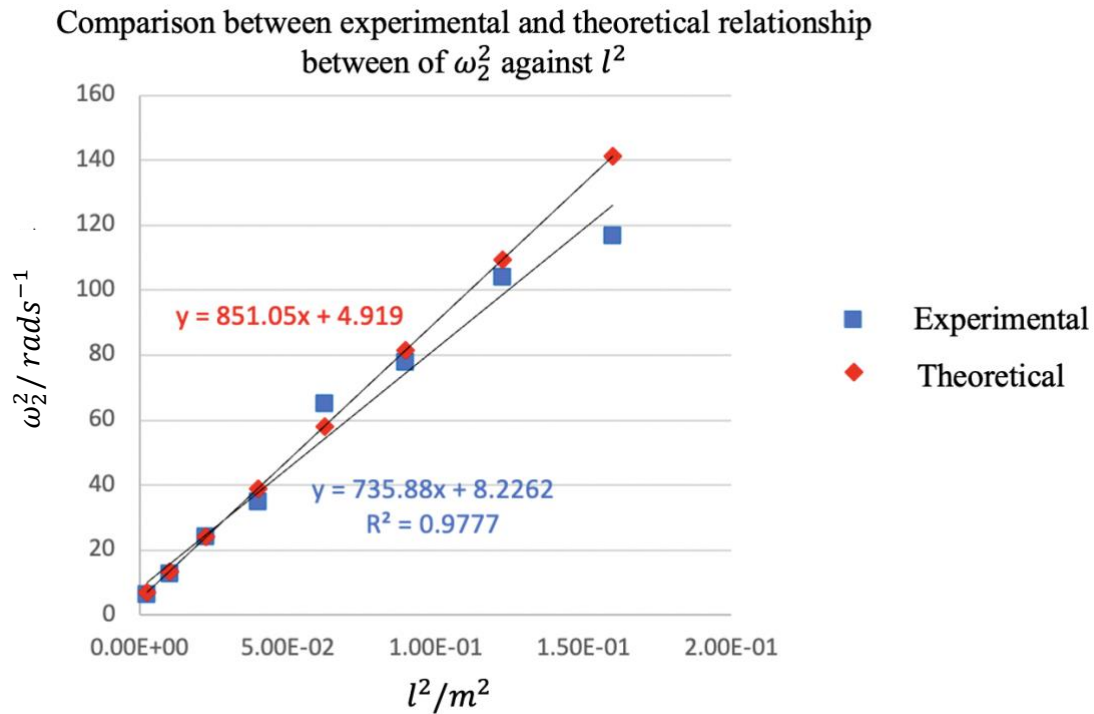
From (25), experimentally, $\omega_1 = (1.898 \pm 0.006) rads^{-1}$, while theoretically, given by

(15), $\omega_1 = (2.218 \pm 0.0007) rads^{-1}$. On the other hand, from (26), $\omega_2^2 = (735 \pm 50) l^2 + (8 \pm 4)$, while theoretically, given by (16), $\omega_2^2 = (850.99 \pm 0.03) l^2 + (4.9 \pm 0.4)$.

The experimental and theoretical graphs of ω_1 against l and that of ω_2^2 against l^2 can be placed on the same axes for comparison.



Graph 6. Comparison between the experimental and theoretical graph of ω_1 against l



Graph 7. Comparison between the experimental and theoretical graph of ω_2^2 against l^2

	Gradient	y-intercept
Theoretical	0.000	2.218
Experimental	0.000	1.898
Percentage error	0.000%	14.43%

Table 13. Comparison in Graph 6.

	Gradient	y-intercept
Theoretical	851.05	4.919
Experimental	735.88	8.226
Percentage error	%13.532	67.23%

Table 14. Comparison in Graph 7.

From **Graph 2**, the line of best fit is a horizontal line, indicating that ω_0 is independent of l , which matches with the theoretical model. All the data points in **Graph 2** lie on the horizontal line of best-fit, indicating a high precision in the results. However, from **Graph 6**., it was shown that $(\omega_1)_{Exp} < (\omega_1)_{Theory}$ for all l , indicating that the experimental data are systematically lower than that of the theoretical ones, demonstrating a low accuracy.

From **Graph 3**, the line of best fit is a straight line that has a positive y-intercept with an R^2 value of 0.9777, showing that ω_2^2 increases linearly with l^2 , which coincides with the theoretical model. However, the data points on **Graph 3** alternate above and below the line of best-fit, showing a relatively low precision in the results. From **Graph 7**, it was shown that generally, $(\omega_2^2)_{Exp} < (\omega_2^2)_{Theory}$ for the same value of l^2 , indicating that the experimental data are systematically lower than that of the theoretical model, indicating a relatively low accuracy.

11.1.3) Final theoretical results

$l^2 \pm \Delta(l^2)/m^2$	$(\omega_{avg} - \frac{1}{2}\sqrt{\frac{Mgl_0}{2I}})^2 / rad^2 s^{-2}$	$\Delta \left[(\omega_{avg} - \frac{1}{2}\sqrt{\frac{Mgl_0}{2I}})^2 \right] / rad^2 s^{-2}$
0.1600 ± 0.0008	35.3	0.2
0.1225 ± 0.0007	27.3	0.2
0.0900 ± 0.0006	20.4	0.1
0.0625 ± 0.0005	14.5	0.1
0.0400 ± 0.0004	9.7	0.1
0.0225 ± 0.0003	6.02	0.07
0.0100 ± 0.0002	3.36	0.05
0.0025 ± 0.0001	1.76	0.03

Table 15. Theoretical data for $(\omega_{avg} - \frac{1}{2}\sqrt{\frac{Mgl_0}{2I}})^2$ and l^2

$l^2 \pm \Delta(l^2)/m^2$	$(\omega_{Beat} + \frac{1}{2}\sqrt{\frac{Mgl_0}{2I}})^2 / rad^2 s^{-2}$	$\Delta \left[(\omega_{Beat} + \frac{1}{2}\sqrt{\frac{Mgl_0}{2I}})^2 \right] / rad^2 s^{-2}$
0.1600 ± 0.0008	35.3	0.2
0.1225 ± 0.0007	27.3	0.2
0.0900 ± 0.0006	20.4	0.1
0.0625 ± 0.0005	14.5	0.1
0.0400 ± 0.0004	9.7	0.1
0.0225 ± 0.0003	6.02	0.07
0.0100 ± 0.0002	3.36	0.05
0.0025 ± 0.0001	1.76	0.03

Table 16. Theoretical data for $(\omega_{Beat} + \frac{1}{2}\sqrt{\frac{Mgl_0}{2I}})^2$ and l^2

11.1.4) Final comparison between experimental and theoretical results

From (29),

$$(\omega_{avg} - \frac{1}{2} \sqrt{\frac{Mgl_0}{2I}})^2 = (172 \pm 11) l^2 + (1.6 \pm 0.9)$$

while given by (27), theoretically,

$$(\omega_{avg} - \frac{1}{2} \sqrt{\frac{Mgl_0}{2I}})^2 = (213.0 \pm 0.1) l^2 + (1.22 \pm 0.01)$$

From (30),

$$(\omega_{Beat} + \frac{1}{2} \sqrt{\frac{Mgl_0}{2I}})^2 = (194 \pm 12) l^2 + (3 \pm 1)$$

While given by (28), theoretically,

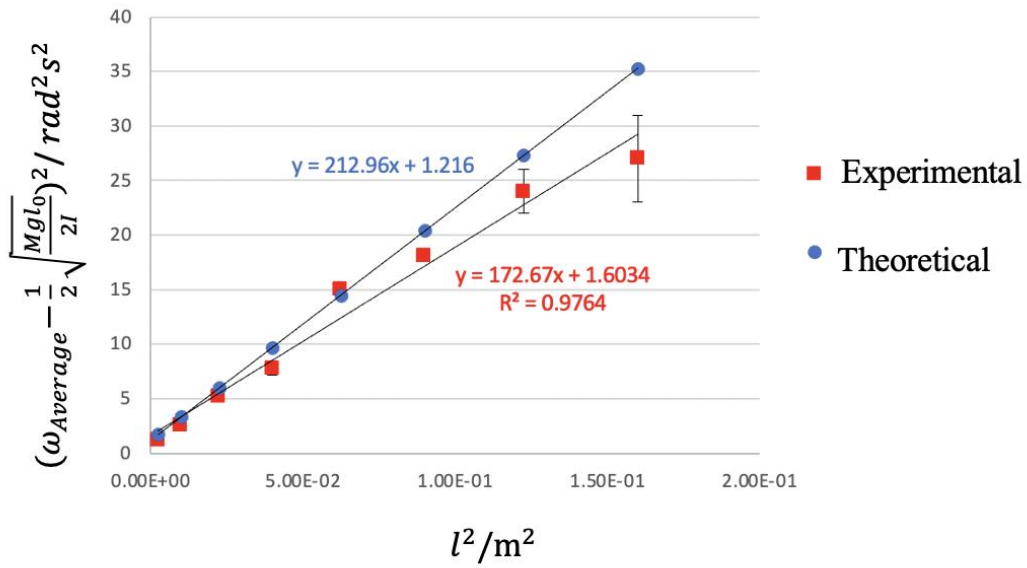
$$(\omega_{Beat} + \frac{1}{2} \sqrt{\frac{Mgl_0}{2I}})^2 = (213.0 \pm 0.1) l^2 + (1.22 \pm 0.01)$$

The Experimental and theoretical graphs of $(\omega_{avg} - \frac{1}{2} \sqrt{\frac{Mgl_0}{2I}})^2$ against l^2 and $(\omega_{Beat} +$

$\frac{1}{2} \sqrt{\frac{Mgl_0}{2I}})^2$ against l^2 can be placed on the same axes for comparison.

Comparison between experimental and theoretical relationship

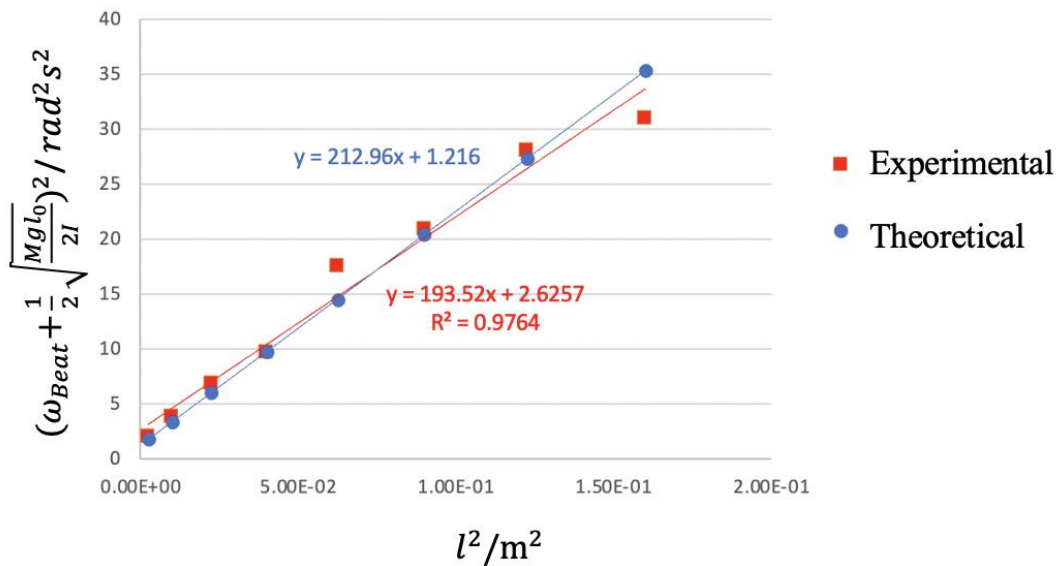
between of $(\omega_{Average} - \frac{1}{2}\sqrt{\frac{Mgl_0}{2I}})^2$ against l^2



Graph 8. Comparison between experimental and theoretical graph of $(\omega_{avg} - \frac{1}{2}\sqrt{\frac{Mgl_0}{2I}})^2$ against l^2

Comparison between experimental and theoretical relationship

between of $(\omega_{Beat} + \frac{1}{2}\sqrt{\frac{Mgl_0}{2I}})^2$ against l^2



Graph 9. Comparison between experimental and theoretical graph of $(\omega_{Beat} + \frac{1}{2}\sqrt{\frac{Mgl_0}{2I}})^2$ against l^2

	Gradient	y-intercept
Theoretical	212.96	1.216
Experimental	172.67	1.6034
Percentage error	18.919%	31.86%

Table 17. Comparison in Graph 8.

	Gradient	y-intercept
Theoretical	212.96	1.216
Experimental	193.52	2.6257
Percentage error	9.1285%	116%

Table 18. Comparison in Graph 9.

From both **Graph 3** and **Graph 4**, a best fit straight line that has a positive y-intercept is applied and shows an R^2 value of 0.9764, showing that both ω_{avg} and ω_{beat} increases with l , which matches with the theoretical model.

In **Graph 3**, the data points again alternate above and below the line of best fit, showing a relatively low precision in the results. Generally, the error bars are of a small magnitude. However, there are exceptions such as when $l^2 = 0.1225\text{m}^2$ and $l^2 = 0.1600\text{m}^2$, which contain a relatively large vertical error bar. In addition, from **Graph 8**, $(\omega_{avg} -$

$$\frac{1}{2}\sqrt{\frac{Mgl_0}{2l}}_{exp}^2 < (\omega_{avg} - \frac{1}{2}\sqrt{\frac{Mgl_0}{2l}})_{theory}^2 \quad \text{for the same } l^2, \text{ showing that the experimental data}$$

are systematically lower than that of the theoretical one, indicating a low accuracy in the results.

In **Graph 4**, the data points again deviate above and below the line of best fit, also indicating the low precision of the results. However, from **Graph 9**, the experimental and theoretical curve are in proximity, showing the low systematic error in between, and a high accuracy in the results.

11.2) Conclusion

The hypothesis is verified. ω_1 remains unchanged with the distance of the spring from the pivot l , ω_2^2 increases linearly with l^2 ; while ω_{avg}^2 and ω_{beat}^2 increases with l . This concludes how an increase in the distance increases the angular frequencies of oscillation. Such results are a classical analogue to how an increase in bond strength increases the angular frequency of oscillation in molecules, executing such phenomenon on a macroscopic scale.

However, despite the verified relationships obtained, there are evidence of existing systematic and random errors in the data collected, which would be further explained in the later session in which different sources of error would be investigated upon and their effects on the results of the experiment will be considered.

12) Evaluation

The systematic errors in this experiment mainly stem from the assumptions made when deriving the theoretical model to explain the relationship between the angular frequencies of oscillation and the length of the spring from the pivot. The assumptions made in the model were small angle approximation, assumption of the same masses of the pendula, and the absence of any external damping forces. Below are the evaluation of such factors and the significance on how they would affect the experimental results systematically, in addition with the random errors, and how such errors can be reduced to give more accurate and precise results respectively.

12.1) Systematic error from large angle oscillation

Small angle approximation was used throughout the theoretical calculations. Such was adopted as the initial angle of release of the pendulum was kept at an angle less than 10° . However, when the experiment is conducted, the angle which the pendulum oscillates is larger than 10° , resulting in significant inaccuracies in small angle approximation to calculate the beat and average frequencies of oscillation. To decrease the discrepancies between the theoretical and experimental angular frequencies, a new model, which includes large angle oscillations, shall be introduced to describe the oscillations of the pendulum. From **Appendix C**, the modified periods for the first and second normal mode of oscillation, which are given by T_1'' and T_2'' respectively are,

$$T_1'' = 2\pi \sqrt{\frac{2I}{Mgl_0}} \left(1 + \frac{1}{4} \sin^2 \frac{\theta_0}{2} + \dots\right)$$
$$T_2'' = 2\pi \sqrt{\frac{2I}{Mgl_0 + 4kl^2}} \left(1 + \frac{1}{4} \sin^2 \frac{\theta_0}{2} + \dots\right)$$

By $\omega = \frac{2\pi}{T}$, the modified 1st and 2nd normal mode angular frequencies are given by,

$$\omega_1'' = \sqrt{\frac{Mgl_0}{2I}} \left(\frac{1}{1 + \frac{1}{4} \sin^2 \frac{\theta_0}{2} + \dots} \right) \quad (31)$$

$$\omega_2'' = \sqrt{\frac{Mgl_0}{2I} + \frac{2kl^2}{I}} \left(\frac{1}{1 + \frac{1}{4} \sin^2 \frac{\theta_0}{2} + \dots} \right) \quad (32)$$

Therefore, the modified average angular frequencies are given by,

$$\omega_{avg}'' = \frac{\omega_1'' + \omega_2''}{2} \quad (33)$$

$$\omega_{avg}'' = \left(\frac{1}{2} \sqrt{\frac{Mgl_0}{2I}} + \frac{1}{2} \sqrt{\frac{Mgl_0}{2I} + \frac{2kl^2}{I}} \right) \left(\frac{1}{1 + \frac{1}{4} \sin^2 \frac{\theta_0}{2} + \dots} \right) \quad (34)$$

As Shown above, $\omega_{average}''$ is lower than $\omega_{average}$ for the same values of l since

$\left(\frac{1}{1 + \frac{1}{4} \sin^2 \frac{\theta_0}{2} + \dots} \right) < 1$, which decreases the discrepancies between the theoretical model and the experimental results (as shown in **Graph 8**.)

To reduce such a systematic error, the pendulum should be carefully displaced from rest without applying any external forces. The initial angle of release should also be kept at a smaller angle (preferably less than 10°) to ensure the accuracy of small angle approximation.

12.2) Systematic error from assuming that the pendula are of the same mass

The masses of the pendulum are assumed to be the same throughout the theoretical calculations. This assumption was originally justified since the mass of both pendula are similar, being $(45.55 \pm 0.01)\text{g}$ and $(46.19 \pm 0.01)\text{g}$, which have a percentage difference of 1.395%. However, the experimental results suggest that such contributes to the systematic errors. As both pendula are assumed to have a mass of $(45.55 \pm 0.01)\text{g}$, the mass of the second pendulum is underestimated. ω_1^2 and ω_2^2 for unequal masses are given by,

$$\omega_1^{2''} = \frac{2gM_1l_0 + 4l^2k + gl_0\alpha - \sqrt{g^2l_0^2\alpha^2 + 16l^4k^2}}{4I} \quad (84)$$

$$\omega_2^{2''} = \frac{2gM_1l_0 + 4l^2k + gl_0\alpha + \sqrt{g^2l_0^2\alpha^2 + 16l^4k^2}}{4I} \quad (85)$$

(Appendix D) where M_1 is the mass of the (45.55 ± 0.01) g pendulum and $\alpha = 6 \times 10^{-4}$ kg is the mass difference between the pendula. By substituting in the parameters into (60) and (62) it is found out that the first and second normal mode, it was found out that $\omega_1^{2''}$ and $\omega_2^{2''}$ takes up a similar value to that of ω_1^2 and ω_2^2 , showing that the assumption that the mass of the pendula are the same is accurate and has insignificant contribution to the systematic error in the experimental results.

12.3) Systematic error from damping

As observed from the angular displacement-time graphs, the separation between successive crests/troughs and beats increases over time, indicating an increase in the period of oscillation, thus, a decrease in angular frequencies. One of the reasons for such a decrease would be due to the presence of external damping forces (e.g., air resistance) acting on the system. The presence of damping forces can be supported by the observation that the amplitude of the sinusoidal wave envelop decreases with time (shown in **Figure 9**.)

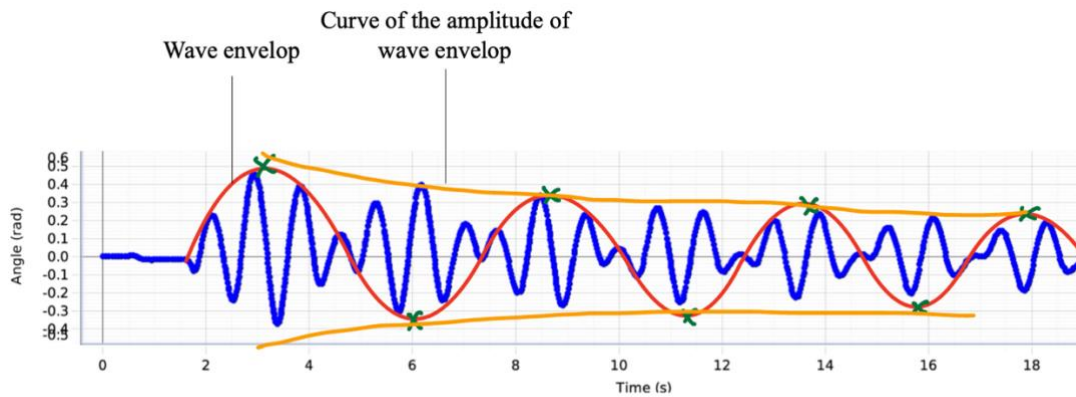


Figure 9. Angular displacement-time graph showing the decrease in amplitude of wave envelop with time

Such resistive forces decrease the angular velocity, and thus, angular frequency of oscillation. Quantitatively, the effect of damping by air resistance on the average and beat angular frequency can be evaluated using a modified Lagrangian $\mathcal{L}''$, given by,⁸

$$\mathcal{L}'' = e^{ct}(T - V) \quad (35)$$

Where c is the damping coefficient, and T and V are the total kinetic and potential energy of the system respectively. From **Appendix E**, when the pendula are subjected to air resistance, the first and second normal mode angular frequencies are given by,

$$\omega_1''^2 = \frac{Mgl_0}{2I} - \left(\frac{c}{2}\right)^2 \quad (95)$$

$$\omega_2''^2 = \left(\frac{Mgl_0}{2I} + \frac{2kl^2}{I}\right) - \left(\frac{c}{2}\right)^2 \quad (102)$$

In which both ω_1'' and ω_2'' are smaller than that of ω_1 and ω_2 , reducing discrepancies between the theoretical model and the experimental results (shown in **Graph 6** and **Graph 7**).

In addition, the modified average angular frequencies ω_{avg}'' is given by,

$$\omega_{avg}'' = \frac{\omega_1'' + \omega_2''}{2}$$

$$\omega_{avg}'' = \frac{1}{2} \left(\sqrt{\frac{Mgl_0}{2I} - \frac{c^2}{4}} + \sqrt{\frac{Mgl_0}{2I} + \frac{2kl^2}{I} - \frac{c^2}{4}} \right) \quad (36)$$

In which both ω_{avg}'' is smaller than ω_{avg} , decreasing the discrepancies between the experimental results and the theoretical model (shown in **Graph 8**).

One way to reduce such error is to use heavier pendula (i.e., half meter rules) which can decrease the rate of damping, which decreases the rate of decrease of the period of oscillation. An alternative method would be to use a pendulum of a shorter thickness to reduce its contact surface area with air molecules when oscillating. Different geometries of

⁸ *Lagrangian mechanics with Friction: Step by Step Guide with example.* (n.d.).
<https://profoundphysics.com/friction-in-lagrangian-mechanics/>

pendula can also be considered (e.g., rods) which could reduce the drag coefficient, reducing the air resistance acting on it. The above methods could reduce the effect of damping on the reduction in the angular frequencies of oscillation.

12.4) Random error from instruments

The instruments used in the investigation have scale error.

The rotary motion sensor could measure and record frequencies with a scale uncertainty of $\pm 0.001\text{Hz}$. The random uncertainty associated with frequency measurements were minimized by taking the average over 5 separate trials for the same l . The error associated with such can also be minimized by using more precise rotary motion sensors for frequency measurement, which are not available in the school laboratory.

The electronic balance also has a scale uncertainty of $\pm 0.01\text{g}$. Such random error can be reduced by using a more precise electronic balance which gives more significant figures.

13) Extension

A possible extension is to alter other parameters, such as the lengths or masses of the pendula, and investigate their respective effect on the angular frequencies of oscillation.

Further investigation can be done on 3-mass coupled pendula, in which a combination of different springs and masses can be used.

Appendix A

The expression for the moment of inertia of the pendulum is required for deriving the theoretical relationship between the angular frequencies of oscillation and the distance of the spring from the pivot. Considering the set-up of a single pendulum below,

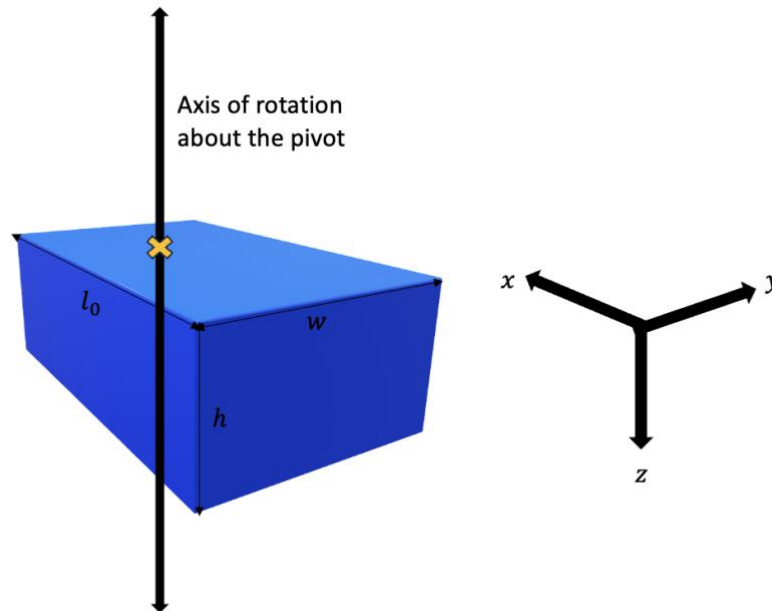


Figure 10. 3D model of a half meter rule

It was assumed that the spring connected to the half meter rule has neglectable mass as compared to it, the diameter of the holes on the ruler are neglectable, and that mass is uniformly distributed about the whole ruler.

A.1) Parallel axis theorem

The parallel axis theorem states that the moment of inertia about any axis parallel to the axis through the center of mass $I_{parallel\ axis}$ is given by,⁹

$$I_{parallel\ axis} = I_{cm} + md^2 \quad (37)$$

Where I_{cm} is the moment of inertia of the body, m is the mass of the body, and d is the distance between the axis of through the center of mass of the body and the axis of rotation.

⁹ Reference: Parallel Axis theorem [Online]. HyperPhysics Concepts <http://hyperphysics.phy-astr.gsu.edu/hbase/parax.html>

A.2) Perpendicular axis theorem

For a planar object, the moment of inertia about an axis perpendicular to the plane is the sum of the moments of inertia of two perpendicular axes through the same point in the plane of the object¹⁰

A.3) Moment of inertia of half meter rule

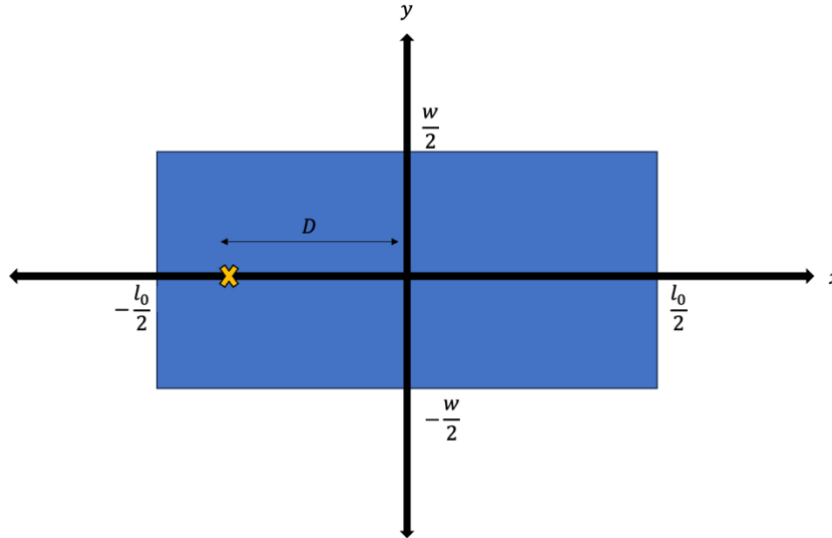


Figure 11. 2D Top view of the half meter rule

Considering the moment of inertia of the half meter rule when it rotates about its center of mass. The equation for the moment of inertia of the wooden rule about the y-axis is given by,

$$I_y = \int_{-\frac{l_0}{2}}^{\frac{l_0}{2}} x^2 dM \quad (38)$$

Taking an infinitesimal volume of cuboid along the y-z plane, the volume of the section is,

$$dV = wh dx \quad (39)$$

Where h and w is the thickness and the width of the half meter rule respectively. The expression for the mass of the half meter rule is thus given by,

$$dM = \rho dV = \rho wh dx \quad (40)$$

¹⁰ Reference: Perpendicular Axis theorem [Online]. HyperPhysics Concepts <http://hyperphysics.phy-astr.gsu.edu/hbase/perpx.html#:~:text=For%20a%20planar%20object%2C%20the,moments%20of%20strictly%20planar%20objects.>

Substituting dM into the (38) and evaluating the integral

$$I_y = \int_{-\frac{l_0}{2}}^{\frac{l_0}{2}} x^2 \rho wh \, dx \quad (41)$$

$$\begin{aligned} &= \rho wh \int_{-\frac{l_0}{2}}^{\frac{l_0}{2}} x^2 \, dx \\ &= \frac{1}{3} \rho wh \left[\left(\frac{l_0}{2} \right)^3 + \left(\frac{l_0}{2} \right)^3 \right] = \frac{2}{3} \rho wh \left[\left(\frac{l_0}{2} \right)^3 \right] \\ I_y &= \frac{1}{12} \rho wh l_0^3 \end{aligned} \quad (42)$$

As the mass of the half meter rule m is given by $m = \rho V = \rho wh l_0$, the moment of inertia of the half meter rule about its center of mass is given by,

$$I_y = \frac{1}{12} M l_0^2 \quad (43)$$

Similarly, its moment of inertia about the x-axis is given by,

$$I_x = \frac{1}{12} M w^2 \quad (44)$$

Where w is the width of the half meter rule. According to the perpendicular axis theorem, the moment of inertia about the z-axis I_z is given by,

$$I_z = I_x + I_y \quad (45)$$

$$I_z = \frac{1}{12} M (w^2 + l_0^2) \quad (46)$$

According to the parallel axis theorem, the moment of inertia of the half meter rule, about the pivot, is given by,

$$I = \frac{1}{12} M (w^2 + l_0^2) + M D^2 \quad (47)$$

In which D is the distance of the pivot from the closer end of the half meter rule.

Appendix B

B.1) Uncertainty of the moment of inertia

Referring to **Appendix A**, the moment of inertia I of a pendulum is given by,

$$I = \frac{1}{12}M(w^2 + l_0^2) + MD^2 \quad (48)$$

The uncertainty of the moment of inertia is given by,

$$\Delta I = \Delta\left(\frac{1}{12}Ml_0^2\right) + \Delta\left(\frac{1}{12}Mw^2\right) + \Delta(MD^2) \quad (49)$$

In which $\Delta\left(\frac{1}{12}Ml_0^2\right)$ is given by,

$$\begin{aligned} \frac{\Delta\left(\frac{1}{12}Ml_0^2\right)}{\frac{1}{12}Ml_0^2} &= \frac{\Delta M}{M} + 2\left(\frac{\Delta l_0}{l_0}\right) \\ \Delta\left(\frac{1}{12}Ml_0^2\right) &= \frac{1}{12}Ml_0^2 \left[\frac{\Delta M}{M} + 2\left(\frac{\Delta l_0}{l_0}\right)\right] \end{aligned} \quad (50)$$

And which $+\Delta\left(\frac{1}{12}Mw^2\right)$ is given by,

$$\begin{aligned} \frac{\Delta\left(\frac{1}{12}Mw^2\right)}{\frac{1}{12}Mw^2} &= \frac{\Delta M}{M} + 2\left(\frac{\Delta w}{w}\right) \\ \Delta\left(\frac{1}{12}Mw^2\right) &= \frac{1}{12}Mw^2 \left[\frac{\Delta M}{M} + 2\left(\frac{\Delta w}{w}\right)\right] \end{aligned} \quad (51)$$

And $\Delta(MD^2)$ is given by,

$$\begin{aligned} \frac{\Delta(MD^2)}{MD^2} &= \frac{\Delta M}{M} + 2\left(\frac{\Delta D}{D}\right) \\ \Delta(MD^2) &= MD^2 \left[\frac{\Delta M}{M} + 2\left(\frac{\Delta D}{D}\right)\right] \end{aligned} \quad (52)$$

Taking the sum of (50), (51) and (52),

$$\Delta I = \frac{1}{12}Ml_0^2 \left[\frac{\Delta M}{M} + 2\left(\frac{\Delta l_0}{l_0}\right)\right] + \frac{1}{12}Mw^2 \left[\frac{\Delta M}{M} + 2\left(\frac{\Delta w}{w}\right)\right] + MD^2 \left[\frac{\Delta M}{M} + 2\left(\frac{\Delta D}{D}\right)\right] \quad (53)$$

B.2) Uncertainty of $(\omega_{avg} - \frac{1}{2}\sqrt{\frac{Mgl_0}{2I}})^2$

$$\frac{\Delta\left[\left(\omega_{avg} - \frac{1}{2}\sqrt{\frac{Mgl_0}{2I}}\right)^2\right]}{\left(\omega_{avg} - \frac{1}{2}\sqrt{\frac{Mgl_0}{2I}}\right)^2} = 2 \left[\frac{\Delta\left(\omega_{avg} - \frac{1}{2}\sqrt{\frac{Mgl_0}{2I}}\right)}{\left(\omega_{avg} - \frac{1}{2}\sqrt{\frac{Mgl_0}{2I}}\right)} \right]$$

$$\Delta \left[\left(\omega_{avg} - \frac{1}{2} \sqrt{\frac{Mgl_0}{2I}} \right)^2 \right] = 2 \left(\omega_{avg} - \frac{1}{2} \sqrt{\frac{Mgl_0}{2I}} \right) \Delta \left(\omega_{avg} - \frac{1}{2} \sqrt{\frac{Mgl_0}{2I}} \right) \quad (54)$$

In which $\Delta \left(\omega_{avg} - \frac{1}{2} \sqrt{\frac{Mgl_0}{2I}} \right)$ is given by,

$$\Delta \left(\omega_{avg} - \frac{1}{2} \sqrt{\frac{Mgl_0}{2I}} \right) = \Delta \omega_{avg} + \Delta \sqrt{\frac{Mgl_0}{2I}} \quad (55)$$

Where $\Delta \sqrt{\frac{Mgl_0}{2I}}$ is given by,

$$\begin{aligned} \frac{\Delta \left(\frac{Mgl_0}{2I} \right)^{\frac{1}{2}}}{\left(\frac{Mgl_0}{2I} \right)^{\frac{1}{2}}} &= \frac{1}{2} \frac{\Delta \left(\frac{Mgl_0}{2I} \right)}{\left(\frac{Mgl_0}{2I} \right)} \\ \Delta \left(\frac{Mgl_0}{2I} \right)^{\frac{1}{2}} &= \frac{1}{2} \sqrt{\frac{Mgl_0}{2I}} \left(\frac{\Delta M}{M} + \frac{\Delta l_0}{l_0} + \frac{\Delta I}{I} \right) \end{aligned} \quad (56)$$

Substituting (49) into (48),

$$\Delta \left(\omega_{avg} - \frac{1}{2} \sqrt{\frac{Mgl_0}{2I}} \right) = \Delta \omega_{avg} + \frac{1}{2} \sqrt{\frac{Mgl_0}{2I}} \left(\frac{\Delta M}{M} + \frac{\Delta l_0}{l_0} + \frac{\Delta I}{I} \right) \quad (57)$$

Substituting (50) into (47) yields,

$$\Delta \left[\left(\omega_{avg} - \frac{1}{2} \sqrt{\frac{Mgl_0}{2I}} \right)^2 \right] = 2 \left(\omega_{avg} - \frac{1}{2} \sqrt{\frac{Mgl_0}{2I}} \right) \left[\Delta \omega_{avg} + \frac{1}{2} \sqrt{\frac{Mgl_0}{2I}} \left(\frac{\Delta M}{M} + \frac{\Delta l_0}{l_0} + \frac{\Delta I}{I} \right) \right] \quad (58)$$

B.3) Uncertainty of $\left(\omega_{Beat} + \frac{1}{2} \sqrt{\frac{Mgl_0}{2I}} \right)^2$

As the uncertainty of additions and subtractions can be found with the same method, the

uncertainty of $\left(\omega_{Beat} + \frac{1}{2} \sqrt{\frac{Mgl_0}{2I}} \right)^2$ can be found with a similar manner as above. Following

the form of (58), $\Delta \left[\left(\omega_{Beat} + \frac{1}{2} \sqrt{\frac{Mgl_0}{2I}} \right)^2 \right]$ is given by,

$$\Delta \left[\left(\omega_{Beat} + \frac{1}{2} \sqrt{\frac{Mgl_0}{2I}} \right)^2 \right] = 2 \left(\omega_{Beat} + \frac{1}{2} \sqrt{\frac{Mgl_0}{2I}} \right) \left[\Delta \omega_{Beat} + \frac{1}{2} \sqrt{\frac{Mgl_0}{2I}} \left(\frac{\Delta M}{M} + \frac{\Delta l_0}{l_0} + \frac{\Delta I}{I} \right) \right] \quad (59)$$

Appendix C

To find the period of the first and second normal mode of oscillation at large initial angles

To calculate the period of the first and second normal mode of oscillation, we may consider setting up equations by utilizing the conservation of energy. To find the period of the first normal mode of oscillation, we consider the total energy of one of the pendulums,

$$E_{Tot} = \frac{Mgl_0}{2}(1 - \cos \theta_0) \quad (60)$$

where θ_0 is the initial angle of release, of which corresponds to the energy when it is at maximum angular displacement. While the energy of the pendulum at an arbitrary moment is given by,

$$E_{arbitrary} = \frac{1}{2}I\omega^2 + \frac{Mgl_0}{2}(1 - \cos \theta) \quad (61)$$

By the law of conservation of energy,

$$E_{Tot} = E_{arbitrary}$$
$$\frac{Mgl_0}{2}(1 - \cos \theta_0) = \frac{1}{2}I\omega^2 + \frac{Mgl_0}{2}(1 - \cos \theta) \quad (62)$$

Substituting $\omega = \frac{d\theta}{dt}$ into (62) and rearranging the terms,

$$Mgl_0(1 - \cos \theta_0) - Mgl_0(1 - \cos \theta) = I \left(\frac{d\theta}{dt} \right)^2$$
$$\frac{Mgl_0}{I}(\cos \theta - \cos \theta_0) = \left(\frac{d\theta}{dt} \right)^2$$
$$\frac{d\theta}{dt} = \sqrt{\frac{Mgl_0}{I}} \sqrt{\cos \theta - \cos \theta_0} \quad (63)$$

By $1 - \cos \theta = 2 \sin^2 \frac{\theta}{2}$,

$$\frac{d\theta}{\sqrt{\sin^2 \frac{\theta_0}{2} - \sin^2 \frac{\theta}{2}}} = 2 \sqrt{\frac{Mgl_0}{2I}} dt$$

$$\frac{d\theta}{(\sin^2 \frac{\theta_0}{2}) \sqrt{1 - \frac{\sin^2 \frac{\theta}{2}}{\sin^2 \frac{\theta_0}{2}}}} = 2 \sqrt{\frac{Mgl_0}{2I}} dt \quad (64)$$

The Ratio $\frac{\sin^2 \frac{\theta}{2}}{\sin^2 \frac{\theta_0}{2}}$ in the square root of the denominator will oscillate (but not with simple

harmonic motion) between -1 and +1, and so we will make the identification $\sin \phi = \frac{\sin^2 \frac{\theta}{2}}{\sin^2 \frac{\theta_0}{2}}$,

and by letting $b = \sin \frac{\theta_0}{2}$, so that

$$\sin \frac{\theta}{2} = b \sin \phi \quad (65)$$

$$\cos \frac{\theta}{2} = \left(1 - \sin^2 \frac{\theta}{2}\right)^{\frac{1}{2}} = (1 - b^2 \sin^2 \phi)^{\frac{1}{2}} \quad (66)$$

Hence, (64) can be rewritten in the form of,

$$\int \frac{d\theta}{b \sqrt{1 - \sin^2 \phi}} = 2 \int \sqrt{\frac{mgl_0}{2I}} dt \quad (67)$$

From differentiating (65), we have

$$\begin{aligned} \frac{d}{d\theta} \left(\sin \frac{\theta}{2} \right) &= \frac{d}{d\theta} (b \sin \phi) \\ \frac{1}{2} \cos \frac{\theta}{2} d\theta &= b \cos \phi d\phi \\ d\theta &= 2b \frac{\sqrt{1 - \sin^2 \phi}}{\sqrt{1 - b^2 \sin^2 \phi}} d\phi \end{aligned} \quad (68)$$

Substituting (68) into (67) gives,

$$\int \frac{2b}{b \sqrt{1 - \sin^2 \phi}} \frac{\sqrt{1 - \sin^2 \phi}}{\sqrt{1 - b^2 \sin^2 \phi}} d\phi = 2 \int \frac{d\phi}{\sqrt{1 - b^2 \sin^2 \phi}} \quad (69)$$

Thus, the integral in (67) becomes,

$$\int \frac{d\phi}{\sqrt{1 - b^2 \sin^2 \phi}} = \int \sqrt{\frac{mgl_0}{2I}} dt \quad (70)$$

Obtaining an elliptical integral of the first kind, we may find the power series solution to this integral by expanding the function $\frac{1}{\sqrt{1 - b^2 \sin^2 \phi}}$,

$$(1 - b^2 \sin^2 \phi)^{-\frac{1}{2}} = 1 + \frac{1}{2}b^2 \sin^2 \phi + \frac{3}{8}b^4 \sin^4 \phi + \dots \quad (71)$$

Therefore, the integral (70) becomes,

$$\int \left(1 + \frac{1}{2}b^2 \sin^2 \phi + \frac{3}{8}b^4 \sin^4 \phi + \dots\right) d\phi = \int \sqrt{\frac{Mgl_0}{2I}} dt \quad (72)$$

Suppose initially at $t = 0, \theta = 0$ such that $\sin \phi = 0$ and $\phi = 0$. As the pendulum oscillate for one period at $t = T_1''$, the pendulum returns to its lowest position, when $\phi = 2\pi$. Inserting such limits into the integral and integrating them term by term,

$$\int_0^{2\pi} \left(1 + \frac{1}{2}b^2 \sin^2 \phi + \frac{3}{8}b^4 \sin^4 \phi + \dots\right) d\phi = \int_0^{T_1''} \sqrt{\frac{Mgl_0}{2I}} dt \quad (73)$$

$$2\pi + \frac{1}{2}\pi \sin^2 \frac{\theta_0}{2} + \dots = T_1'' \sqrt{\frac{Mgl_0}{2I}}$$

$$T_1'' = 2\pi \sqrt{\frac{2I}{Mgl_0}} \left(1 + \frac{1}{4}\sin^2 \frac{\theta_0}{2} + \dots\right) \quad (74)$$

Applying the same arguments for the 2nd normal mode of oscillation, the period becomes,

$$T_2'' = 2\pi \sqrt{\frac{2I}{Mgl_0 + 4kl^2}} \left(1 + \frac{1}{4}\sin^2 \frac{\theta_0}{2} + \dots\right) \quad (75)$$

Appendix D

D.1) 1st and 2nd normal mode angular frequency for unequal masses

When the masses of the 2 pendula are different, the equations of motion are,

$$\ddot{\theta} + \frac{l^2 k}{I}(\theta - \phi) + \frac{M_1 g l_0}{2I}\theta = 0 \quad (76)$$

$$\ddot{\phi} - \frac{l^2 k}{I}(\theta - \phi) + \frac{M_2 g l_0}{2I}\phi = 0 \quad (77)$$

Where M_1 and M_2 are the $(45.55 \pm 0.01)\text{g}$ and $(46.19 \pm 0.01)\text{g}$ pendulum respectively.

Expressing (76) and (77) in matrix representation,

$$\begin{bmatrix} \frac{d^2}{dt^2} + \frac{l^2 k}{I} + \frac{M_1 g l_0}{2I} & -\frac{l^2 k}{I} \\ -\frac{l^2 k}{I} & \frac{d^2}{dt^2} + \frac{l^2 k}{I} + \frac{M_2 g l_0}{2I} \end{bmatrix} \begin{bmatrix} \theta \\ \phi \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad (78)$$

Defining the θ and ϕ in terms of complex constants and assume an oscillatory solution of the form $e^{i\omega t}$,¹²

$$\begin{bmatrix} \theta \\ \phi \end{bmatrix} = \Re \begin{bmatrix} X \\ Y \end{bmatrix} e^{i\omega t} \quad (79)$$

Substituting (79) to form the eigenfunction equation,

$$\begin{bmatrix} -\omega^2 + \frac{l^2 k}{I} + \frac{M_1 g l_0}{2I} & -\frac{l^2 k}{I} \\ -\frac{l^2 k}{I} & -\omega^2 + \frac{l^2 k}{I} + \frac{M_2 g l_0}{2I} \end{bmatrix} \begin{bmatrix} X \\ Y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad (80)$$

Setting the deterministic equation to 0,

$$\begin{vmatrix} -\omega^2 + \frac{l^2 k}{I} + \frac{M_1 g l_0}{2I} & -\frac{l^2 k}{I} \\ -\frac{l^2 k}{I} & -\omega^2 + \frac{l^2 k}{I} + \frac{M_2 g l_0}{2I} \end{vmatrix} = 0 \quad (81)$$

¹² Jarvis M. (2016) *Waves and normal mode*. University of Oxford Department of Physics

$$\left(-\omega^2 + \frac{l^2 k}{I} + \frac{M_1 g l_0}{2I}\right) \left(-\omega^2 + \frac{l^2 k}{I} + \frac{M_2 g l_0}{2I}\right) - \left(\frac{l^2 k}{I}\right)^2 = 0$$

$$\omega^2 = \frac{1}{4I} (M_1 g l_0 + M_2 g l_0 + 4l^2 k) \pm \frac{1}{4I} \sqrt{M_2^2 g^2 l_0^2 + M_1^2 g^2 l_0^2 - 2g^2 M_1 M_2 l_0^2 + 16l^4 k^2} \quad (82)$$

Given that $M_2 > M_1$, let $M_2 = M_1 + \alpha$, where α is the mass difference,

$$\omega^2 = \frac{2gM_1 l_0 + 4l^2 k + g l_0 \alpha \pm \sqrt{g^2 l_0^2 \alpha^2 + 16l^4 k^2}}{4I} \quad (83)$$

In which ω_1^2 and ω_2^2 are given by,

$$\omega_1^{2''} = \frac{2gM_1 l_0 + 4l^2 k + g l_0 \alpha - \sqrt{g^2 l_0^2 \alpha^2 + 16l^4 k^2}}{4I} \quad (84)$$

$$\omega_2^{2''} = \frac{2gM_1 l_0 + 4l^2 k + g l_0 \alpha + \sqrt{g^2 l_0^2 \alpha^2 + 16l^4 k^2}}{4I} \quad (85)$$

Appendix E

E.1) Modified 1st normal mode angular frequency under damping

The total kinetic energy of the system T is given by,

$$T = \frac{1}{2}I \left(\frac{d\theta}{dt} \right)^2 + \frac{1}{2}I \left(\frac{d\theta}{dt} \right)^2 = I \left(\frac{d\theta}{dt} \right)^2 \quad (85)$$

The total potential energy of the system is given by,

$$V = -\frac{Mgl_0 \cos \theta}{2} - \frac{Mgl_0 \cos \theta}{2} = -Mgl_0 \cos \theta \quad (86)$$

Therefore, the modified Lagrangian of the system is given by,

$$\mathcal{L}'' = e^{ct}(T - V) \quad (35)$$

$$\mathcal{L}'' = e^{ct} \left[I \left(\frac{d\theta}{dt} \right)^2 + Mgl_0 \cos \theta \right] \quad (87)$$

When θ is very small, $\cos \theta \approx 1 - \frac{\theta^2}{2}$, such that,

$$\mathcal{L}'' = e^{ct} \left[I \left(\frac{d\theta}{dt} \right)^2 + Mgl_0 \left(1 - \frac{\theta^2}{2} \right) \right] \quad (88)$$

Applying the Euler-Lagrange equation to $\mathcal{L}''$,

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}''}{\partial \dot{\theta}} \right) = \frac{\partial \mathcal{L}''}{\partial \theta} \quad (89)$$

where,

$$\frac{\partial \mathcal{L}''}{\partial \theta} = e^{ct}(-Mgl_0\theta) \quad (90)$$

$$\frac{\partial \mathcal{L}''}{\partial \dot{\theta}} = e^{ct} \left(2I \frac{d\theta}{dt} \right) \quad (91)$$

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}''}{\partial \dot{\theta}} \right) = \frac{d}{dt} e^{ct} \left(2I \frac{d\theta}{dt} \right) = 2Ie^{ct} \left(\frac{d^2\theta}{dt^2} + c \frac{d\theta}{dt} \right) \quad (92)$$

Combining (90), (91) and (92),

$$\begin{aligned}
 2Ie^{ct} \left(\frac{d^2\theta}{dt^2} + c \frac{d\theta}{dt} \right) &= e^{ct} (-Mgl_0\theta) \\
 2I \left(\frac{d^2\theta}{dt^2} + c \frac{d\theta}{dt} \right) &= -Mgl_0\theta \\
 \frac{d^2\theta}{dt^2} + c \frac{d\theta}{dt} + \frac{Mgl_0}{2I} \theta &= 0
 \end{aligned} \tag{93}$$

Noticing that this second-ordered differential equations take the same form as the one for damped harmonic motion, the general solution can be given by,

$$\theta = A_0 \exp\left(-\frac{ct}{2}\right) \cos \omega_0'' t \tag{94}$$

¹³where A_0 is the initial amplitude of oscillation, and ω_0'' is the modified frequency after damping. As ω_0'' is given by,

$$\begin{aligned}
 \omega_0''^2 &= \omega_0^2 - \left(\frac{c}{2}\right)^2 \text{ }^{14} \\
 \omega_0''^2 &= \frac{Mgl_0}{2I} - \left(\frac{c}{2}\right)^2
 \end{aligned} \tag{95}$$

E.2) Modified 2nd normal mode angular frequency under damping

Applying the same arguments as shown from E.1, the modified Lagrangian thus becomes,

$$\mathcal{L}'' = e^{ct} \left[I \left(\frac{d\theta}{dt} \right)^2 + Mgl_0 \left(1 - \frac{\theta^2}{2} \right) - 2kl^2 \theta^2 \right] \tag{96}$$

¹³ Reference: Ford, A. L. Young, H. D. Freedman, R. A. *Sears and Zemansky's University Physics with Modern Physics* (13th edition). Pearson

¹⁴ Reference: Ford, A. L. Young, H. D. Freedman, R. A. *Sears and Zemansky's University Physics with Modern Physics* (13th edition). Pearson

Applying the Euler-Lagrange equation on $\mathcal{L}''$,

$$\frac{\partial \mathcal{L}''}{\partial \theta} = e^{ct}(-Mgl_0\theta - 4kl^2\theta) \quad (97)$$

$$\frac{\partial \mathcal{L}''}{\partial \dot{\theta}} = e^{ct}\left(2I\frac{d\theta}{dt}\right) \quad (98)$$

$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}''}{\partial \dot{\theta}}\right) = \frac{d}{dt}e^{ct}\left(2I\frac{d\theta}{dt}\right) = 2Ie^{ct}\left(\frac{d^2\theta}{dt^2} + c\frac{d\theta}{dt}\right) \quad (99)$$

Combining (97), (98) and (99),

$$\begin{aligned} 2Ie^{ct}\left(\frac{d^2\theta}{dt^2} + c\frac{d\theta}{dt}\right) &= e^{ct}(-Mgl_0\theta - 4kl^2\theta) \\ 2I\left(\frac{d^2\theta}{dt^2} + c\frac{d\theta}{dt}\right) &= -Mgl_0\theta - 4kl^2\theta \\ \frac{d^2\theta}{dt^2} + c\frac{d\theta}{dt} + \left(\frac{Mgl_0}{2I} + \frac{2kl^2}{I}\right)\theta &= 0 \end{aligned} \quad (100)$$

Again, the general solution to the equation is,

$$\theta = A_0 \exp\left(-\frac{ct}{2}\right) \cos \omega_1'' t \quad (101)$$

where,

$$\begin{aligned} \omega_1''^2 &= \omega_1^2 - \left(\frac{c}{2}\right)^2 \\ \omega_1''^2 &= \left(\frac{Mgl_0}{2I} + \frac{2kl^2}{I}\right) - \left(\frac{c}{2}\right)^2 \end{aligned} \quad (102)$$

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